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Open Geospatial Consortium

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I ABSTRACT

GeoSPARQL provides a standard framework for storing, querying and representing geospatial information using RDF and SPARQL. Although GeoSPARQL 1.1 has some support for 3D geometries, its current vocabulary, geometry representations, and query functions provide limited support for 3D spatial data and functions. At the same time, the use of 3D spatial data is increasing across domains such as geospatial information, Building Information Modelling (BIM), construction, infrastructure, emergency response, cultural heritage, and computer graphics.

This white paper investigates the requirements and opportunities for extending GeoSPARQL with improved 3D capabilities. It examines the current capabilities and limitations of GeoSPARQL 1.1, reviews related standards, data formats ontologies, and analyses use cases gathered from different domains. Particular attention is given to the representation and querying of 3D geometries and topology, semantic decomposition of complex 3D objects, interoperability with existing 3D standards, and functions in three-dimensional environments.

The paper evaluates potential extensions according to their relevance, benefits, and technical feasibility. The objective is not to replace existing 3D modelling or geometry standards, but to investigate how GeoSPARQL can provide a common semantic and spatial query layer over heterogeneous 3D data. Such capabilities could enable machine-readable and interoperable 3D spatial analysis without requiring data from different domains to be converted into a single homogeneous representation. Potential benefits include improved integration of GIS, BIM, and other 3D data sources; spatial and semantic querying of complex 3D objects; 3D topology and connectivity analysis; route and accessibility analysis; and applications in construction, safety, emergency response, and other domains involving three-dimensional environments.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

OGC; GeoSPARQL; 3D; 3D geospatial data; 3D BIM; semantic 3D modelling; 3D spatial querying; 3D geometry; 3D topology; RDF; SPARQL; BIM; IFC; CityGML; semantic interoperability; 3D routing;

PREFACE

This white paper has been developed to investigate these questions and to identify potential directions for extending GeoSPARQL with improved 3D capabilities. It brings together requirements and use cases from different domains and examines existing standards, formats, ontologies, and implementations that may inform future development.

The paper is intended as a basis for discussion within the OGC community and among current and prospective GeoSPARQL users and implementers. It does not propose a final specification, but instead identifies potential requirements, approaches, and areas for further investigation. The intention is to support future work towards interoperable 3D spatial querying while building on, rather than replacing, existing 3D standards and technologies.

IV SECURITY CONSIDERATIONS

3D spatial data published as Linked Data may expose sensitive information about buildings, infrastructure, or access routes. Publishers should therefore consider the intended openness of the data and apply appropriate access controls where required. GeoSPARQL itself does not define access-control mechanisms.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- Geonovum
- KurrawongAI
- i3mainz
- Universidade de Lisboa
- Triply
- Graphwise/Ontotext
- AEC Geeks
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1 SCOPE

1 SCOPE

This white paper addresses the potential extension of GeoSPARQL to support 3D spatial modelling, querying, and analysis. It also adresses the integration with existing 3D standards and formats, such as IFC, CityGML, and glTF; Also the benefits across domains including GIS, BIM, construction, infrastructure, emergency response, and cultural heritage are inside the scope. Existing 2D GeoSPARQL capabilities are considered where they are relevant to extending those capabilities to 3D, but improvements to 2D functionality in their own right are outside the scope of this paper.

2 CONFORMANCE

2 CONFORMANCE

This document is an informative OGC White Paper and does not define conformance classes or conformance requirements.

3 NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

There are no normative references in this document.

The following documents contain provisions that are referenced by this document and are necessary for its understanding and application:

4 TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. Geodimension

This is the number of perpendicalar directions in which a geometry extends. A point, for example, extends in no direction, so its topological dimension is 0. A line has a length, but no width or height. Its topological dimension is 1. A cube has a topological dimension of 3.

4.2. coordinatedimension

In the geometric model that GeoSPARQL uses, points are the basic building blocks of geometry. A point geometry can have a different number of coordinates, depending on the space in which it is placed. A point on a flat map, or on the surface of a sphere, needs two coordinates. The *coordinate dimension* is the number of coordinates in the points that define a Geometry.

4.3. spatialDimension

In order to support linear referencing, geometries in GeoSPARQL can have a measure value. This is a relative position along a line or a curve, and does not extend the Geometry in space. The *spatial dimension*, therefore, can be used to denote the number of coordinates in a point, excluding the measurement value.

5 CONVENTIONS

5 CONVENTIONS

RDF classes, properties, and functions are represented using prefixed names. For example, geo:Geometry denotes a GeoSPARQL class, geo:hasGeometry a GeoSPARQL property, and geof:distance a GeoSPARQL function. Geometry examples are expressed using established encodings such as WKT where appropriate.

6

INTRODUCTION

6 INTRODUCTION

OGC GeoSPARQL is an open standard that enables storage and exchange of spatial data on the Web, based on the Resource Description Framework (RDF). In 2020 the OGC white paper Benefits of Representing Spatial Data Using Semantic and Graph Technologies was published. It explained the benefits of GeoSPARQL and described the ways in which GeoSPARQL version 1.0 could be improved. In the meantime, GeoSPARQL version 1.1 has been published, having many improvements described in aforementioned white paper, see also this introduction to GeoSPARQL 1.1 [45].

Although not specifically bound, GeoSPARQL 1.1 is mainly geared towards spatial objects having zero to two dimensions. In other words, things that can be displayed on a flat surface. However, for three-dimensional spatial objects both supply and demand are increasing.

Two new versions of GeoSPARQL are in development: 1.2 and 1.3. GeoSPARQL 1.2 will be an adaption of GeoSPARQL 1.1 which enables GeoSPARQL to be published as an ISO standard. Version 1.2 will have no substantive changes with respect to version 1.1. GeoSPARQL 1.3 is expected to have extended capabilities supporting three-dimensional (3D) space. Not only would that make GeoSPARQL more useful for 3D geospatial data, it would also help in industries and knowledge domains that are not mainly focused on geospatial data, like Building Information Modelling (BIM) and Computer Graphics (CG). This paper describes which extensions to GeoSPARQL for 3D are possible, which are most desirable, and how they could be achieved.

This paper consists of three main parts. The first section describes current capabilities of GeoSPARQL: in its current state, GeoSPARQL does allow modelling three-dimensional objects, but with relevant limitations. The second section describes requirements for extended 3D capabilities in GeoSPARQL. This is based on a market consultation, which resulted in a collection of use cases for additional 3D capabilities. By analysing those use cases, as well as current developments within and outside the OGC, insight in the most needed extensions to GeoSPARQL can be obtained. Requirements are then weighed against feasibility; some extensions are easier to achieve than others. The third and final section describes the expected benefits for additional 3D capabilities in GeoSPARQL for different domains and different roles.

This paper should be interesting for the following audiences:

- Current and prospective users of GeoSPARQL;
- Current and prospective implementers of GeoSPARQL;
- Members of related OGC working groups.

A deep understanding of the Semantic Web and graph technologies is not required for readers of this white paper, but some interest in those topics is assumed, as well as familiarity with the aforementioned white paper Benefits of Representing Spatial Data Using Semantic and Graph Technologies. This paper focuses on specifically on 3D capabilities of GeoSPARQL, not its general benefits.

7

MOTIVATION

7 MOTIVATION

Why do we want 3D semantic modeling? The white paper: “OGC Benefits of Representing Spatial Data Using Semantic and Graph Technologies” describes the beneficiaries and benefits of representing data, including geospatial data, using semantic and graph technologies Abhayaratna2020wp. This section describes the motivation of adding 3D functionality to the GeoSPARQL standard.

While humans can often resolve 3D spatial questions such as “Is cube A inside cylinder B?” — even when the underlying data comes from different systems like GIS and BIM — this typically relies on implicit understanding, conventions, or direct communication between the data creators. For datasets to be meaningfully aligned, the people who created them often need to clarify assumptions, data structures, coordinate systems, or modeling intentions. Without such human-to-human explanation, machines struggle to interpret spatial relationships across heterogeneous sources. A conversion process is required, in which data is made homogeneous for analysis through extraction, transformation, and loading. Based on a homogeneous dataset, conclusions can then be drawn. Performing conversions may result in geometric simplifications, which can limit or even eliminate the functionality of spatial queries Biljecki2019. If one desires digital, automated processing or analysis of 3D spatial questions without requiring conversion, spatial information must be structured in an explicitly machine-readable format. When this is done, initiatives based on federated systems with heterogeneous datasets can obtain computer-generated answers to spatial 3D queries. This requires semantic alignment and standardized geometry processing in machine-driven analysis. In other words, a common vocabulary is needed to semantically define the 3D objects and its 3D sptatial function.

Imagine a constructor who wants to use real-time sensor data at a construction site in combination with modelled 3D data and 3D base registration data. By combining this the person wants to visualize the site, perform accurate spatial checks and calculations, and automatically update machine instructions for safe, efficient and autonomous execution”

The system uses 3D GeoSPARQL to integrate the data of the 3D design model, the 3D base registration model and 3D real-time sensor data to:

- Spatially associate sensor data with objects in the design and registration model
- Calculate the permissible tolerances
- Detect the deviations
- Dynamic adjust the design model
- Instruct the machine with the changes
- Visualize the new situation

For both humans and machines, it must be unambiguous what is meant by the queries that are needed to do the job and the resulting answers. It should not matter whether the data is provided in a 3D-GIS, 3D-BIM or 3D-CG standard vocabulary.

A geo-spatial 3D specification would support data consumers by:

- access and understanding of 3D-geometry and 3D-topology from multiple sources;
- the integration of 3D-data geospatial and non-geospatial;
- data quality evaluation of the given 3D-data;
- Geometric and topological description in natural language.

A geo-spatial 3D specification would support data providers by:

- Maximizing the use of 3D-geometry and 3D-topology data;
- Defining multiple 3D topology or geometries;
- Assuring the quality of 3D geospatial data.

8

CURRENT CAPABILITIES

8 CURRENT CAPABILITIES

Realizing the motivation described above requires technical and semantic support for the standardized representation, integration, and querying of 3D spatial information. GeoSPARQL could provide such a foundation. First it is necessary to understand the current capabilities of the standard. This provides the basis for considering where additional capbilities or extensions may be needed.

8.1. GeoSPARQL

GeoSPARQL is the most common geospatial extension of SPARQL. It was accepted as an OGC standard in 2012 and revised as GeoSPARQL 1.1 in 2024.

According to the standard document, “The OGC GeoSPARQL standard supports representing and querying geospatial data on the Semantic Web. GeoSPARQL defines a vocabulary for representing geospatial data in RDF, and it defines an extension to the SPARQL query language for processing geospatial data”.

8.1.1. Requirements addressed

In order to define which capabilities GeoSPARQL needs to adopt for full 3D compatibility, we first take a look at GeoSPARQL 1.1 current capabilities with regards to 3D.

8.1.1.1. Dimensionality

In this white paper, when we discuss three dimensional data, we mean three dimensional geometries; instances of the class geo:Geometry. GeoSPARQL 1.1 defines three different properties for geometry that have to do with dimensionality:

1. geo:dimension, or *topological dimension*: This is the number of perpendicalar directions in which a geometry extends. A point, for example, extends in no direction, so its topological dimension is 0. A line has a length, but no width or height. Its topological dimension is 1. A cube has a topological dimension of 3.

2. geo:coordinateDimension: In the geometric model that GeoSPARQL uses, points are the basic building blocks of geometry. A point geometry can have a different number of coordinates, depending on the space in which it is placed. A point on a flat map, or on the surface of a sphere, needs two coordinates. The *coordinate dimension* is the number of coordinates in the points that define a Geometry.

3. geo:spatialDimension: In order to support linear referencing, geometries in GeoSPARQL can have a measure value. This is a relative position along a line

or a curve, and does not extend the Geometry in space. The *spatial dimension*, therefore, can be used to denote the number of coordinates in a point, excluding the measurement value.

The extended capabilities for GeoSPARQL discussed in this white paper concern the cases where the *spatial dimension* (wich can be expressed with property geo:spatialDimension) of geometry has the value 3.

The table below shows the values of the three dimension properties for some Geometries (expressed as Well Known Text (WKT)).

Table 1

GEOMETRY (WKT)	GEO:DIMENSION	GEO:COORDINATEDIMENSION	GEO:SPATIALDIMENSION
point (4 15)	0	2	2
point Z (4 15 3)	0	3	3
linestring (4 15, 13 2)	1	2	2
point M (4 15 60)	0	3	2
point ZM (4 15 3 60)	0	4	3

8.1.1.2. Vocabulary

GeoSPARQL version 1.1 consists of a Core module, a Topology Vocabulary extension, a Geometry extension, a Geometry Topology extension, an RDFS Entailment extension, and a Query Rewrite extension. Each of these modules will be briefly explained below.

8.1.1.3. Core Module

GeoSPARQL version 1.1 includes a CORE module. This defines the basic classes, relationships, and literals. These classes are suitable for both 2D and 3D.

8.1.1.3.1. Topology Vocabulary Extension

This extension provides standard definitions for spatial relationships that can exist between one spatial object and another.

Relations between geometries have been defined using three different sets of rules:

— Simple Features Relations

— Egenhofer Relations

— Region Connection Calculus RCC8

These topological definitions specify topological relations in a 2D-context.

8.1.1.3.2. Geometry Extension

GeoSPARQL 1.1 geometry module defines a class `Geometry` as a subclass of `SpatialObject`.

An instance of `Geometry` is not restricted to two dimensions.

A fine-grained classification of `Geometry` can use the Simple Feature Vocabulary which extends the class `Geometry` with further types, such as *Point*, *Polygon* etc.

The Simple Features vocabulary allows for the definition of 3D variants of:

— (Multi)Points

— (Multi)LineStrings

— (Multi)Polygons

— Polyhedralsurface

It does not include commons 3D primitives, such as *Cube* or *Mesh* surfaces which are integral parts of 3D representations. Nor does it include extruded geometry.

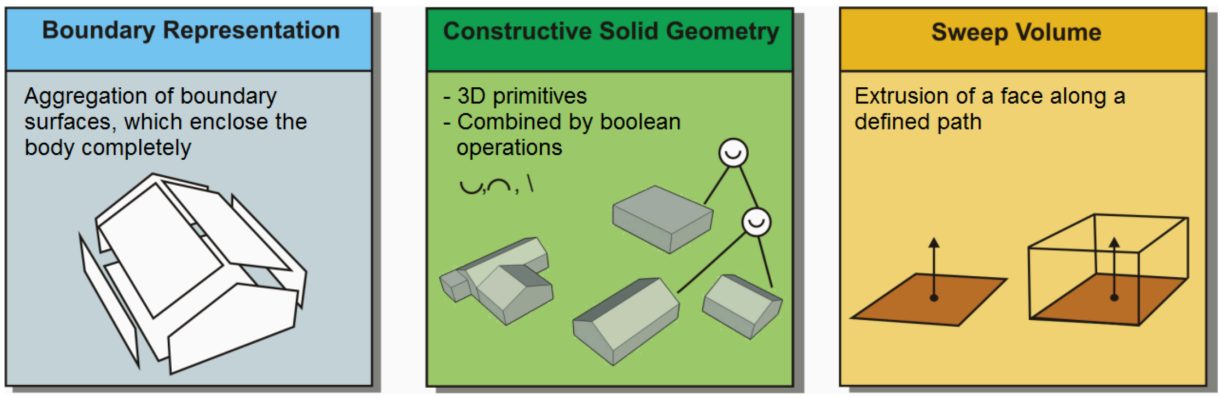


Figure 1 — Possible approaches for representing 3D objects in IFC

Concerning metadata of 3D models, GeoSPARQL 1.1 provides properties which can be reused in 3D contexts. In particular, the properties are:

— `geo:hasVolume`

— `geo:hasMetricVolume`

Further 3D-related metadata properties such as projection matrices are not part of the current GeoSPARQL 1.1 standard.

A first requirement for 3D support is the ability to save 3D data in a knowledge graph.

The Geometry extension enables geometries to be defined within the GeoSPARQL framework. Supported geometry formats include RDF literals in WKT, GML, GeoJSON, and KML.

These literals are investigated for the storage of 3D data.

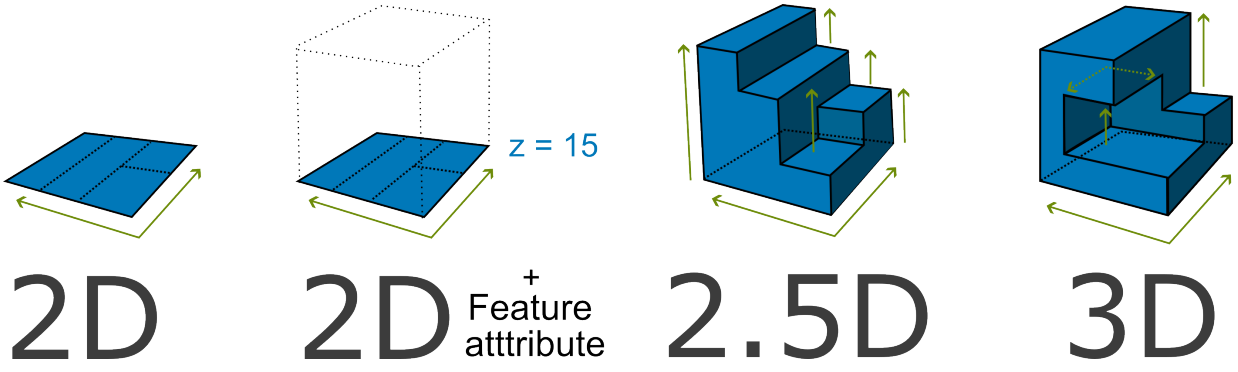


Figure 2 — Difference between 2D, 2D with height attribute 2.5D and 3D geometry

Table 2

Literal Type	Z-Coordinate Supported	Z-Coordinate Supported + height attribute	2.5D	3D
WKT Literal	Yes	Yes	Yes	Yes
GML Literal	Yes	Yes	Yes	Only with extension Schema
KML Literal	Yes	Yes	Yes	As import from COLLADA
GeoJSON Literal	Yes	Yes	Yes	Yes
DGGS Literal				

GeoSPARQL 1.1 also does not restrict the usage of coordinate reference systems with 3D support. There are currently almost 300 coordinate reference systems in the database epsg.io which can be used to describe 3D data encoded in the GeoSPARQL graph literals listed above.

GeoSPARQL allready supports linking multiple geometries (e.g., 2D and 2.5D) to a single feature. We aim to extend this with 3D geometries.

For example, a tree in the real world may be represented in databases with both a 2D and a 3D geometry.

This should not require duplication of the tree entity — there can be one tree instance with both geometries linked, avoiding redundancy in the data model.

8.1.1.4. Query functions with 3D support

GeoSPARQL 1.1 functions currently do not offer fully-featured 3D support. However, there are functions which may take into account the Z coordinate, if they are available.

Table 3

GeoSPARQL function	Z-Coordinate Supported	2.5D	3D
geof:is3D	Yes	Yes	Yes
geof:minZ	Yes	Yes	Yes
geof:maxZ	Yes	Yes	Yes

These functions check for the presence of Z coordinates or filter out maximum and minimum Z coordinates of the given geometry.

8.1.1.5. Geometry Topology Extension

Another extension is the Geometry Topology extension. This provides query functions that return relationships between different geometries based on the topology vocabulary extension.

8.1.1.6. RDFS Entailment Extension

The RDFS Entailment extension provides rules for reasoning over geometries. Based on specific statements, additional information can be inferred. This kind of logical inference can also be applied to geometry. GeoSPARQL includes logic for reasoning over simple features geometries.

8.1.1.7. Query Rewrite Extension

GeoSPARQL allows queries such as whether “Feature A” is located within “Feature B” using its vocabulary. The Query Rewrite extension specifies a RIF rule that enables query rewriting. However, this extension does not support the rewriting of 3D queries.

8.1.2. Adoption of GeoSPARQL 1.1

The article “A GeoSPARQL Compliance Benchmark”, Jovanovik, Homburg and Spasić [44] describes the results of a GeoSPARQL compliance benchmark. The measure of the extent of

GeoSPARQL support in GeoSPARQL-enabled RDF triplestores was done in 2021. The article concludes that the GeoSPARQL standard, almost nine years after its initial release, is often only partially supported by major triplestore vendors. This raises the question of whether adding functionality to a standard increases implementation complexity, or whether new and more powerful capabilities can instead stimulate adoption by enabling new and more valuable use cases for software developers and users.

On the OGC Implementation Database there is no organisation compliant with the OGC GeoSPARQL standard. There are four organisations that are implementing the GeoSPARQL (v 1.0) standard.

8.1.3. GeoSPARQL Aligments

GeoSPARQL is aligned with a range of established ontologies and vocabularies covering different aspects of geospatial information, metadata, provenance, time, sensors, and linked data. See Annex E of the GeoSPARQL Standard. Alignments with additional ontologies may be relevant for a broader 3D context. As an example, ontologies such as Reloc, BOT and ifcOWL can be interesting.

8.1.4. GeoSPARQL functionality and geometries comparison

GeoSPARQL functionality can be compared with established spatial database extensions such as PostGIS H2GIS and SpatiaLite. The alligment offers a useful basis for comparing common spatial operations and capabilities with those defined by GeoSPARQL. The selection for spatial databases is not intended to be exhaustive. Other spatial databases, libraries and systems could also be included, depending on the scope of the comparison. The selected systems should therefore be regarded as representative reference implementations rather than a complete overview of available spatial technologies.

The comparison with existing open-source spatial technologies is intended to identify gaps between the functionality currently provided by GeoSPARQL and the capabilities available in established spatial databases. This provides an indication of the development potential of GeoSPARQL. The comparison can therefore be used to identify potential areas for further development and to assess which capabilities could add value to future versions of the standard.

When focusing specifically on 3D functionality, the largest differences in the comparison are found in geometry creation and editing, spatial measurements, and operations related to triangulation and topographic geometry. Also the topological functions shows some gaps between GeoSPARQL and the functionality in the spatial databases. These areas contain functionality available in the compared spatial technologies that is not represented in GeoSPARQL. They therefore provide the clearest areas for investigating potential further development of 3D functionality in GeoSPARQL. See Annex A: Alignment GeoSPARQL and Open Source Relational Databases with Geospatial Support

Besides the available functions, the geometry representations used by GeoSPARQL, such as WKT and GML, can also be compared with geometry representations used in other standards and technologies, such as IFC, STEP and Open CASCADE. The comparison shows that the largest differences in 3D geometry representation occur beyond basic points, curves

and surfaces. IFC, STEP and OpenCascade provide more extensive representations for solids, volumes, boolean and constructive geometry, swept and revolved geometry and 3D transformations and spatial operations. These areas therefore provide relevant starting points for investigating potential extensions of GeoSPARQL for more comprehensive 3D geometry representation and processing. See Annex B: Allignment of Geometry

9

RELATED WORK

9 RELATED WORK

Various vocabularies, standards, and domain-specific approaches provide concepts for describing 3D geometries, topology, spatial relationships, and their semantics. This section reviews these related developments to understand the different ways in which 3D spatial information is currently modeled and to identify concepts and approaches that may complement or inform the further development of 3D in GeoSPARQL.

A substantial body of work shows that **there is no single universal 3D web ontology**, but several reusable, often domain- or format-specific vocabularies to describe geometry, topology, scenes, and domain semantics. Two principal flavours of ontologies can be identified in existing literature:i) those meant for the generic representation of 3D objects and shapes, and ii) a growing body of domain-specific ontologies [31].

This section starts by providing a brief overview of these two flavours of ontologies. It then dives into domain-specific initiatives, considering Geo-spatial, Cultural Heritage and BIM. This section closes by enumerating various encoding formats relevant for 3D.

9.1. 3D Exchange formats

A new GeoSPARQL standard should be compatible with existing approaches to encode 3D data. Semantic technologies should enable users to encode important metadata in the knowledge graph, while retaining the possibility to use already well-known 3D geometry formats either as String literals or file references. bonduel2019including shows how common widespread geometry formats could be included into knowledge graphs, which could be one starting point to investigate the integration within GeoSPARQL.

9.1.1. Standard for the Exchange of Product model data (STEP) and OntoSTEP

STEP (Standard for the Exchange of Product model data) is an ISO standard for representing and exchanging product and engineering information, including CAD models. OntoSTEP translates STEP concepts into an ontology-based representation to support semantic integration and reasoning. OntoSTEP is a research-oriented approach that translates STEP/EXPRESS schemas and instances into OWL to enable semantic integration, querying, and reasoning. Initially developed at NIST, it has been further investigated through later implementations, but it remains a research approach rather than an established standard or widely adopted 3D ontology.

- OWL encodings of STEP AP203/AP214/AP242 and BREP/CSG concepts for CAD models [43].

— Other proposals link geometric models to spatial properties or count non-manifold features [FlotynskiWalczak2017] [43].

9.1.2. STereoLithography (STL)

STL files describe only the surface geometry of a three-dimensional object without any representation of color, texture or other common CAD model attributes. The STL format specifies both ASCII and binary representations and was invented by the Albert Consulting Group for 3D Systems in 1987. Nowadays it is mainly used for 3D printing purposes, which makes is a plausible candidate for support in GeoSPARQL 3D settings of movable objects such as in cultural heritage. Since the STL format does not contain a geospatial reference, the reference needs to be provided externally if STL files were to be used in a geospatial context. However, STL objects could be used in local CRS contexts as well.

9.1.3. Wavefront OBJ Format (OBJ)

The OBJ format is a file format developed by Wavefront Technologies for the Advanced 3D Visualizer animation package. Its roots therefore lie in the world of 3D animation. The format represents vertex coordinates, texture coordinates, as well as vertex normals, parameter space vertices and faces of a 3D model. These are all elements which would need to be considered and supported in an upcoming GeoSPARQL 3D release.

9.1.4. Polygon File Format (PLY)

The Polygon File Format is a format which was designed to store 3D data from 3D scanning environments, which makes it especially widespread in the Cultural Heritage application area. The format consists of a vertex list and a faces list with optional parameters per vertex. Like the OBJ format, supporting PLY would imply including support for vertices and faces of a 3D model and their relationships in a knowledge graph.

9.2. Geometry metadata and linking

9.2.1. Geometry Metadata Ontology (GOM)

The Geometry Metadata Ontology (GOM) introduces classes for surfaces which are common in the 3D world, but uncommon for 3D geometries, for instance Meshes or NURBS surfaces. The ontology can serve as an inspiration for extensions in GeoSPARQL 3D and may or may not be fully or partly adopted in the process of standardization.

9.2.2. Ontology for Managing Geometry (OMG)

The Ontology for Managing Geometry (OMG) describes states and derivation of geometries which may be useful to the description of data in GeoSPARQL 3D. For example. an extrusion in 3D from a 2D geometry template would have a derivation relation from the 2D geometry which one might want to see modeled in GeoSPARQL 3D.

9.2.3. File Ontology for Geometry Formats (FOG)

The File Ontology for Geometry Formats (FOG) provides a bridge between ontology descriptions of geometries and data file descriptions of 3D contents. Since many 3D datasets are of considerable size, their inclusion into the knowledge graph as String literals might hinder the query performance of said knowledge graph. It therefore stands to reason whether to adopt the same or a similar method of geometry access that FOG provides for GeoSPARQL 3D.

9.3. Generic 3D representation and semantic models

9.3.1. X3D Ontology

Extensible 3D (X3D) Graphics is an open standard to declaratively representing interactive 3D computer graphics for publishing, viewing, printing and archiving 3D models on the Web. X3D was eventually approved by ISO (ISO/IEC 19775-19777) and become an computer graphics industry standard. It was first proposed in 1997 and has since been subject to various updates. More recently it has been developed towards the Semantic Web, in order to gain interoperability and enable intelligent 3D operations and reasoning. Development has been hosted by the Web3D Consortium.

Currently in its version 4, the X3D Ontology maps the full X3D standard (X3DUOM) into OWL 2, providing classes and properties for nodes, geometry, appearance, animation, and scene graph structure, enabling query and reasoning over X3D scenes [FlotynskiWalczak2017] georgieva2023transforming [41] [39] flotynski2020recent [_975906]. X3D is regarded as a comprehensive and formally grounded 3D ontology, adding fundamental concepts and roles of the 3D modelling industry not originally covered by X3D.

The X3D Ontology has also been used to transform non-semantic 3D file formats into RDF with languages such as RML or XSLT georgieva2023transforming [39].

The X3D format feichtenhofer2020×3d, brutzman2010×3d, X3DOM behr2009x3dom and its Ontology version brutzman2020×3d provide a data format and a blueprint for how 3D data is handled in computer vision approaches. X3D is an interesting format to investigate for 3D components which need to be modeled within a GeoSPARQL 3D specification. It is a likely candidate for a literal format and might be integrated using interlinking approaches to

the already existing X3D vocabulary. However, since X3D was invented with computer vision approaches in mind it does not define any geospatial aspects that could be considered by a GeoSPARQL 3D implementation.

9.3.2. VRML / Java3D spatial ontologies

Application ontologies abstracting VRML & Java3D node types plus a separate spatial ontology for topological, directional and metric relations pacheco2018creating.

9.3.3. Common Shape Ontology

– represents manifold/non-manifold shapes, point sets, hierarchical model groups, positions, orientations, keyframe animation [FlotynskiWalczak2017].

9.4. Domain-specific 3D ontologies

9.4.1. Geospatial and 3D City Model (CityGML)

CityGML and OntoCityGML

CityGML [koble2005citygml] is an open standardized data model for the exchange of 3D models of cities and landscapes. Since its standardization by OGC, CityGML has seen a wide variety of proposed extensions and alignments to other data standards.

CityJSON and CityRDF

[46] proposed the extension of the CityGML and its accompanying serialization CityJSON as an ontology model. Currently, the CityRDF standardization group works towards the standardization of this CityGML data model in RDF.

The proposed CityRDF model, even though focused on CityGML and 3D building contents, contains valuable data descriptions which could be generalized in a GeoSPARQL 3D approach. One such element is the description of the appearance of a 3D-modeled building using colors or textures, so-called surface materials. In addition, CityGML includes relations of 3D primitives between each other, which could serve as an inspiration for similary-named functions or properties in the GeoSPARQL 3D ontology model.

[46] describes the CityGML Ontology, which is an ontology model to formalize the CityGML standard towards a representation in OWL. It shows a workflow of identification, classification and selection of the relevant data points needed to describe 3D city models in OWL as derived from building and city models. In addition, it introduces the concept of Level of Information Need, defining the granularity of the information on the various levels of the building and the types of information retrieval — direct/indirect, i.e. information which is commonly obtained

directly from a dataset or using a calculation or reasoning approach. The paper exemplifies the concept in a study case for property unit cost and indoor daylight. The contribution outlined here became the basis of the CityGML ontology which can serve as one broad application area for GeoSPARQL 3D.

Following on this publication [35] presents a system architecture and a set of interfaces that can represent city modeling approaches using dynamic geospatial knowledge graphs. The paper proves the viability of a transition between a SQL based system for 3D building management to a SPARQL based system using a SQL2SPARQL Transformer component. It therefore proves the feasibility of employing linked open data technologies for 3D building data and in particular the CityGML model in practice.

[50] introduces CityGML 3.0 with a newly added Core module including a new space concept and Level Of Detail concept. The given publication sets the standard especially for Level of Detail in 3D buildings, which may be considered for adoption as GeoSPARQL 3D concepts too. In addition, CityGML 3.0 proposes the addition of 3D point clouds for the representation of 3D goemetires of physical spaces and space boundaries. A mapping from IFC to CityGML 3.0 is also proposed by these authors.

CityGML 3.0 therefore represents many of the elements that a GeoSPARQL 3D ontology would need to implement on a more abstract level and can give ideas about future query functions to be implemented in the GeoSPARQL query language.

In addition, a master thesis hansson2024citygml proved the feasibility of not only converting CityGML to a knowledge graph representation, but also its application in triple stores such as Ontop or 3DCityDB. The mapping of CityGML to the knowledge graph was tested using a set of SPARQL queries retrieving different components of building parts. While the thesis could construct a knowledge graph representation from CityGML, it points out that the incorporation of sptaial operations is left to future work and could provide a possible field for GeoSPARQL 3D adoption.

Finally, [30] proposed OntoCityGML, an extension go the CityGML ontolog https://www.sciencedirect.com/science/article/pii/S2666546821000574?via%3Dihub Semantic 3D City Database — An enabler for a dynamic geospatial knowledge graph

Other details and history of Onto CityGML.

CityRDF working group https://github.com/ogcincubator/cityrdf

9.4.2. Cultural Heritage

In the research domain of cultural heritage, 3D models of either cultural heritage artifacts (possibly georeferenced), 3D models of archaeological sites or simply 3D models of ancient buildings are becoming increasinly common.

Building-conservation ontology for 3D annotations. The conservation and restoration of historical monuments require the integration of information from many different disciplines and techniques. This study presents a pipeline that combines semantic, spatial, and morphological data into one 3D information model. The approach was tested on Saint Maurice Church in

Caromb, France, allowing conservation experts to better visualize, correlate, and retrieve information about materials, damage, and other conservation observations. [33];

CIDOC CRM–based models for 3D CH objects and provenance. This paper proposes an ontology-based approach for managing and using 3D content without requiring domain experts to have programming or 3D modelling skills. The actual 3D elements can be created by graphic designers and then linked to domain-specific concepts by developers. Ontologies and automated reasoning can subsequently determine which 3D elements are relevant based on the specified domain information, while semantic queries allow the content to be explored and analysed. The focus is therefore on connecting 3D graphics with domain knowledge rather than on creating the 3D models themselves. walczak2019inference amico2106ontological .

Use Cases

Use Cases in the Cultural Heritage domain include, but are not limited to, the following main interests:

Visualization: The visualization of 3D models for the presentation of such 3D models, for example in the context of a museum. 3D models may be styled with a particular set of textures or modeled with a specific set of colours to highlight certain relevant aspects. The visualization of 3D models is currently standardized by the IIIF 3D working group [48], which aims to create viewing parameter descriptions that 3D viewers may implement, similar to the specifications of IIIF 2D for images.

Object Annotation: 3D models are seen as the subject of a research question in absence of the original artifact for political, practical or other restrictive reasons. Out of all known methods of the representation of cultural heritage artifacts, 3D models provide the most detail when being delivered as a digital artifact and are therefore very often preferred in a research context. Researchers mark noteworthy aspects of the cultural artifact as 3D annotations [47] which may include surface descriptions, volumes of the 3D model or 3D models which are created and placed adjacent to the to-be-annotated 3D model [29].

Relation of Objects: Objects of a specific collections always exist in a spatio-temporal context. It is important to relate these representations via meaningful relations, so that relevant objects of a collection can be retrieved more easily

AI Applications in Cultural Heritage: Usage for annotated areas on 3D models or their derivations for machine learning classifications Stotzner_2023_ICCV 10.2312/gch.20231157

Knowledge Graphs as Metadata descriptions: With the advent of more 3D models being published, the relevance of their creation parameters [51], their contents and their object metadata increases for the usecases of filtering them and also for the possibly automated selection of suitable cultural heritage metadata for e.g. machine learning classifications. Currently, many metadata standards fulfil parts of the description chain and a unified vocabulary to described data types seems to be missing.

Research applications making use of 3D models in Cultural Heritage

This section discusses research projects with 3D contents based on the technologies they use as elaborated in the previous section.

9.4.2.1. 3D models of cuneiform tablets

Cuneiform tablets from ancient Iran provide an interesting research area, since they combine a 3D artifact with textual imprints that are of interest to a variety of research communities including Assyriologsts, Digital Humanists, Computational Linguists and last but not least Computer Scientists. The creation of 3D models of cuneiform tablets provides the best accessibility to the specificities of the original artifact in its absence and 3D scans have been used by computer scientists as the basis for certain machine learning application tasks, even though to this day only as a provision for 2D renderings of their surfaces. Interests of the research community include the description of interesting features such as cuneiform signs on cuneiform tablet surfaces and their connection to other text contents or other cuneiform artifacts. Knowledge graphs provide machine learning approaches Stotzner_2023_ICCV stotzner2023r with unique opportunities to connect different kinds of data using a unified data format.

To describe 3D meshes, several vocabularies have been developed in the context of the cuneiform studies project:

- MeshSPARQL: A vocabulary to describe essential mesh elements
- Gigamesh Metadata Vocabulary: A vocabulary which describes metadata of a 3D model. The metadata can be generated using the Gigamesh Software Framework
- 3DCAP Vocabularies: An ontology model to describe the creation of a 3D model. It has been applied to different scanning softwares

9.4.2.2. CARARE Metadata Schema

The CARARE metadata schema d2013carare is a harvesting schema intended for delivering metadata about an organisation's online collections, heritage assets and their digital resources. The strength of the schema lies with its ability to support the full range of descriptive information about monuments, building, landscape areas and their representations. The CARARE metadata schema builds on existing standards and best practice from a number of different countries in Europe and the rest of the world.

9.4.2.3. IIIF 3D

The International Image Interoperability Framework (IIIF) standard snydman2015international is currently developing a specification for the standardization of 3D viewers. The viewer specification includes the visualization of 3D models including lighting positions, a scene graph and annotations in the form of additional 3D objects added to a 3D scene. A GeoSPARQL 3D standard might need to be compatible with, or include a compatibility statement of, converting 3D objects encoded in GeoSPARQL to the IIIF viewer specification. A prototype implementation might be necessary to comply with this specification.

9.4.3. ChgOnto

Grottoes & heritage ontology. This paper focuses on enhancing 3D models of Chinese grottoes by adding semantic information. It develops a knowledge graph that links 3D elements—such as caves, Buddha statues, walls, and decorations—to their properties and spatial relationships, making the 3D models more meaningful, understandable, and useful for cultural heritage documentation and analysis.for grotto components, spatial relations, with GeoSPARQL + CIDOC CRM yang2023knowledge.

9.4.4. OntPreHer3D

OntPreHer3D is an extension of the CIDOC CRM ontology. It is designed to support the FAIR preservation and documentation of 3D cultural heritage models. The ontology captures relationships between 3D models, their components, and real-world counterparts, while also documenting the reasoning, uncertainties, and decisions behind hypothetical reconstructions. A case study of the Speyer Synagogue demonstrates how differences in modelling processes can affect final 3D models, highlighting the importance of detailed documentation for the long-term preservation, reproducibility, and reuse of digital cultural heritage. OntPreHer3D: Ontology for Preservation of Cultural Heritage 3D Models

9.4.5. BIM and Buildings

9.4.5.1. Industry Foundation Classes (IFC)

BIM is a paradigm in which object-model definitions — with machine-interpretable semantics — are exchanged, rather than relying on CAD drawings that convey only graphical semantics. The predominant open exchange standard is Industry Foundation Classes (IFC). **IFC is unique in the sense that it allows expressing both geometrical as well as non-geometrical semantics.**

Non-geometrical: taxonomy of objects (building elements, relational constructs, scheduling information) and related properties

Geometrical: appromximately 250 constructs related to geometry; even standards with only a geometric focus such as geosparql or collada do not reach that breadth in definitions

IFC has its idiosyncrasies and challenges, but is **unrivaled in terms of market adoption in a highly fragmented industry**.

9.4.5.1.1. Product model

In IFC, a construction work is decomposed into a set of products. These products can have **multiple representations**. For example, a wall can be described as a solid body as well as a two-dimensional axis. These representations facilitate different views on the same data: an editable line segment or an easily visualized volume. The Object-relational nature of the IFC EXPRESS

schema allows intricate relationships such as a representation context that communicates additional intent for the representation or presentation styles that can be granularly assigned to individual faces.

At the same time, such a product separates the **placement** (an hierarchical transformation) from the actual geometry definition. The consequence of this is that in spite of its object-relational nature, IFC product representations cannot be used for building-level topological relationships between solids, because even if two solids are touching in 3D, the fact the the placement is externalized out of the geometry definition (or the fact that faces are constructed procedurally and do not exist explicitly), means that the two faces cannot be opposite oriented twins. As such, relational geometric constructs such as space boundaries are provided as additional supplementary geometries. (Identifying faces in procedural geometries is also challenging (toplogical naming problem), so using geometric semantic overlays to annotate potentially procedural geometries makes sense)

In principle, the IFC schema has been designed in a modular fashion with independent modules for, for example, geometry, materials and meta-data. However in other cases, **semantics and geometry are intertwined** such as tapered extrusions (lofts) where the begin and end profile of a duct carry important semantics.

IFC also allows for **decomposition**, where a whole is aggregated into multiple parts for richer semantics. This allows for example to connect materials and meta-data to the frame and the glazing separately, while still being able to identify the aggregate as a single window. This is not used as frequently, partially due to inability to efficiently instantiate such aggregates as geometry instances.

9.4.5.1.2. Evolving views on geometry

IFC is heavily influenced by the ISO 10303 (STEP) family of standards, but over time adopted its own geometric paradigms:

- Procedural geometry and boolean operations became less prominent with the adoption of ReferenceView in IFC4. Tessellated geometry definitions were added for more compact exchange.
- **Infrastructure definitions** were added with precise mathematical transition curves and a composition of a horizontal, vertical and cant (inclination) profile.
- IFC5 with an **explicit** (most likely triangulated) geometry schema at the core, with semantic overlays to encode the same procedural semantics as a non-mandatory or use-case specific layer. Heavily inspired by USD with layer-based composition for collaborative exchange.

Especially the handling of **tolerances** means that the standard cannot effectively prescribe a consistent outcome in all cases. Tolerances are needed for BRep model with non-linear underlying geometry and/or fixed precision coordinate values, e.g higher degree nurbs curves are typically intersected with numerical approximation, so a vertex that connects two of such curves needs to have seen as a sphere with the local tolerance as its radius. This tolerance is also applied to boolean operations: an subtraction volume can be slightly inwards of the first operand

but is still expected to pierce through the volume and increase surface genus. This contrasts with the desire of using IFC as a legal basis in contracts. NB Tolerances stand in the way of using existing approaches for SFA geometry predicates such as PostGIS+SFCGAL which is based on arbitrary precision boolean logic as implemented in CGAL without tolerances.

9.4.5.1.3. Use cases

The most successful use case on BIM data is **coordination and visualization** where multiple aspect models are geometrically overlaid in order to find issues, which are then communicated to the authoring software where they are addressed. This approach works, because it respects that individual disciplines all have their own specialistic software.

Design to design workflows are much harder to realize, although some Model View Definitions have been developed on top of IFC that enable the transfer of design intent in specific and constrained scenarios, such as precast concrete and structural steel.

Long-term preservation of building information is difficult because of the fact that IFC models are difficult to mutate, because they are so explicit and don't contain the vendor-specific design intelligence. Therefore native software cannot always re-import IFC models, but also the native models degrade over time because of the need to migrate to newer editions of the software. Software that can directly operate on IFC to make modifications is still experimental.

BIM-GIS integration is challenging because it requires familiarity with both domains on where to draw the line between euclidean and non-euclidean geometries and acceptable error metrics.

Simulations on IFC building models are often challenging because the 'bag of individual elements' does not provide a good foundation higher order topological representations required for flow-of-energy type of simulations. For e.g thermal simulation a topological view of space boundaries is required. They have been added as secondary set of ternary relationships, but usage of more specific-purpose and simpler schemas sees still more usage in industry. In general, IFC models are created for a specific purpose and wide-spread usage of those models in nieghbouring domaisn remain challenging because modelling for those neighbouring purposes requires alignment on the worldviews and levels of detail that is often beyond the scope in which such models are procured.

9.4.5.1.4. Implications and questions:

- Euclidean / non-euclidean; is a CRS required?
- Separate representation+placement → enables efficient reinstantiation, but hinders topological relationships because you require the pair of placement+geometry to locate in space
- Geometry as leaf-values or object-relational model : cannot encapsulate geometry into a single literal, but allows for richer semantics
- BRep model (topology + geometry + orientation + location) vs polyhedral model (e.g halfedge) vs explicit loops of point coordinates

— Procedural vs implicit (e.g constraints) vs explicit (polyhedra)

— Tolerances

— Decomposition inside or outside of the ‘geometry ontology’

— Are infra geometries (hor + ver alignment + cant, for positioning and sweeps) in scope?

9.4.5.2. BOT (Building Topology Ontology)

AEC — Building domain ontology for storeys, spaces, elements + 3D mesh links; used with separate geometry vocabularies [36]. |

9.4.5.3. Linked Building Server

Over the last years, Building Information Modelling (BIM) has been increasingly adopted in building projects for its benefits to collaboration and associated risk mitigation. Recent work is trying to make this data available over the web (web-based BIM applications). Most notable in making building data web-ready is the linked building data (LBD) initiative. This group has worked on a number of vocabularies and ontologies, thereby starting from a linked data approach (RDF graphs), aiming at representing building data on the web. Linked Building Data Server

9.4.6. BIMSPARQL

This initiative proposes extending SPARQL with functions for querying IFC building data to simplify queries and derive information from 3D geometry for BIM requirement checking. The functions are represented as RDF vocabularies and implemented using a combination of declarative rules and procedural programming. Evaluation with realistic requirement-checking scenarios demonstrates that the approach supports spatial and logical reasoning and can combine building and regulatory data from multiple sources. [42]

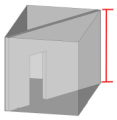
Function	Illustration	Description
pdt:hasBodyGeometry		Returns the geometry form of a product represented as a WKT literal (see Section 4.5). It either retrieves a WKT literal (see Section 4.5) to represent a 3D triangulated surface (TIN Z), or a geometry collection (GeometryCollection Z) in WKT.
pdt:hasAABB		Returns the axis-aligned bounding box of a product as a WKT literal (see Section 4.5).
pdt:hasMVBB		Returns the oriented minimum volume bounding box of a product as a WKT literal (see Section 4.5).
pdt:hasOverallHeight		Returns the height of axis aligned bounding box of a product as a numerical value.
pdt:hasSurface		Returns all plain surfaces of a product. Each of the surfaces is generated as a new binding for the triple pattern which uses this function.
pdt:hasUpperSurface		Returns the upper surface of a product, which is defined as surfaces that have the highest elevation and have normals of nearly (0,0,1), represented as a WKT literal (see Section 4.5). A use case of it is shown in Listing 8.
pdt:hasVolume		Returns the volume of the product as a numerical value.

Figure 3 — IMSPARQL.

9.4.7. stSPARQL

In Modeling and Querying Metadata in the Semantic Sensor Web: the Model stRDF and the Query Language stSPARQL [49], Koubarakis and Kyzirakos describe the extension of SPARQL with temporal information. These paterns can also be used for GeoSPARQL.

9.4.8. OntoBREP and OntoSTEP

OntoBREP is an ontology that can be used to describe CAD by representing boundary representations (B-Rep) using OWL. Geometric entities are represented semantically. For example, a circle is described by its coordinate frame and radius instead of a set of polygons. The ontology represents geometric constraints between objects and their individual faces or edges, enabling CAD geometry to be linked with additional knowledge and used for applications such as shape recognition and robot task planning. [38]

OntoSTEP translates STEP schemas and their instances, originally defined using EXPRESS, into OWL. This creates a semantically representation of product and CAD information. The

approach enables reasoning, inference, and querying over existing CAD data and provides a way to combine information represented using different modeling languages. ontostep

OntoBREP and OntoSTEP represent approaches in which geometry are brought into the Semantic Web at a relatively deep semantic level. This results in potentially large and complex knowledge graphs.

Different approaches for representing geometry in the Semantic Web have been investigated [34] ranging from integrating geometric structures into the knowledge graph to linking semantic resources to external geometry representations or files

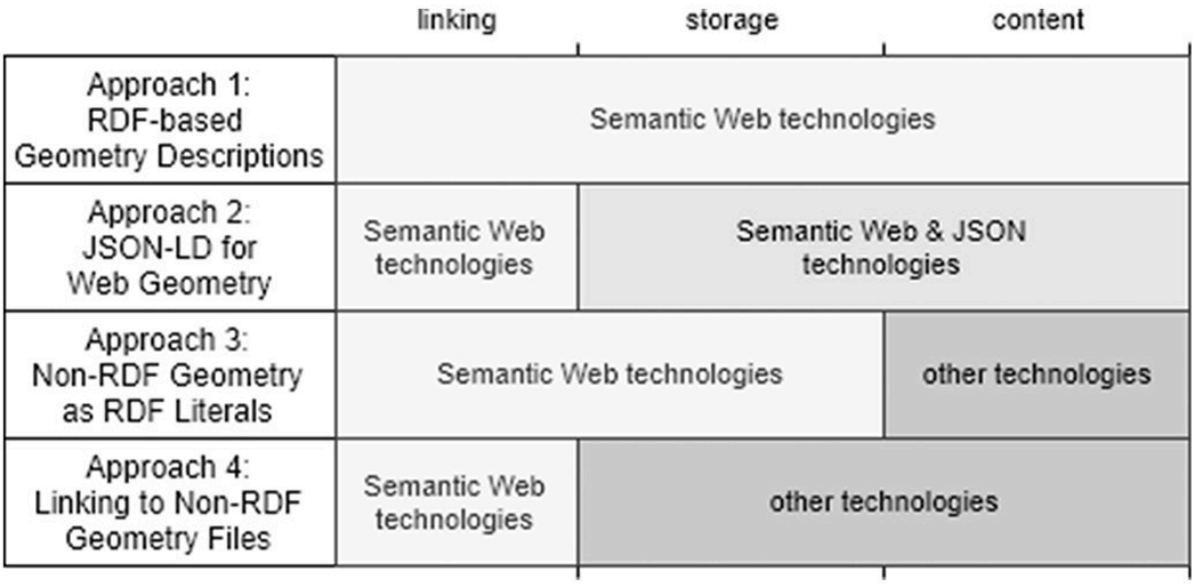


Figure 4 — Approaches_of_Geometry_in_Linked_Data.

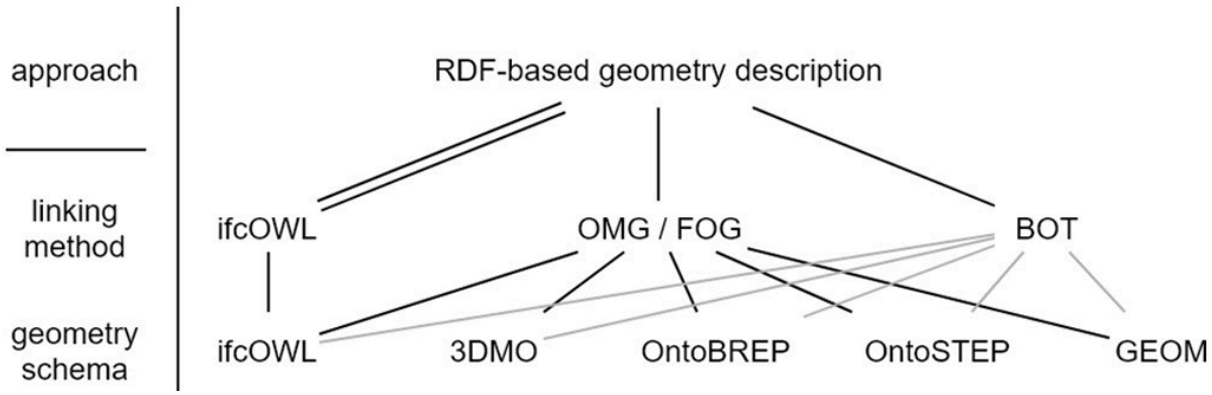


Figure 5 — Implementation_of_approach_1.

9.4.9. ENVON

Robotics / manufacturing ontology that integrates contextual, topologic, geometric info for 3D environments; BFO-based 3D shape ontology; BREP/CSG distinctions [43].

Use of domain specific ontologies Domain-specific 3D ontologies are typically expressed using **RDF/OWL**, defining concepts and relationships that capture the semantics of a particular application domain [FlotynskiWalczak2017] pittarello2006semantic walczak2019inference.

Geometry, topology, scene structure, and domain semantics are often represented by separate but complementary ontologies that can be combined at query time for integrated queries, such as BOT with geometry and sensor (SOSA/SSN or SAREF) ontologies in building applications [43] [36].

9.4.10. Location ontologies

9.4.10.1. Relative Location Ontology (Reloc)

The Relative Location Ontology (RELOC) enables the expression of the spatial relationship between two objects using directional-topological terms. The ontology is designed to be compatible with the natural language-based approach of describing relative locations, enabling the conversion of existing location descriptions into structured spatial graph representations. The ontology provides directional, axis-related properties (e.g., longitudinal axis: front, center, rear) and topological properties (e.g., meet, contained in, intersect). These properties are finally merged into combined properties (e.g., containedInFront) to reflect the spatial information of statements like 'A is in the front part of B'. The application context of the RELOC Ontology is the processing of existing damage documentation of bridges.

9.4.10.2. InfraspatialOT

The InfraspatialOT ontology, described in the paper: infraspatialOT: An Ontology for the Representation of Spatial Relationships in Road Infrastructure, provides a semantic model for representing road infrastructure elements and their spatial relationships to reference elements. It distinguishes between localisation, which describes where an infrastructure element is located along a reference element, and positioning, which captures its relative spatial position, including horizontal and vertical relationships. Positioning relations can further specify whether a distance is involved. This structure enables road infrastructure elements and their spatial relationships to be represented and queried consistently using semantic technologies such as SPARQL.

9.4.11. Medicine and Chmistry

9.4.11.1. 3D Microscopy Metadata Standards (3D-MMS)

The 3D Microscopy Metadata Standards (3D-MMS) was developed by the BRAIN Initiative to establish a consistent way of describing 3D microscopy datasets, particularly datasets of brain tissue. Its purpose is to provide enough information for someone who did not generate the data to understand, find, interpret, reuse, and potentially integrate the dataset with other datasets. The 3D-MMS-metadata can be used to describe how the image volume is oriented and scaled, so that a position inside the microscopy dataset has meaningful spatial coordinates. [52]

9.5. 3D Topology

9.5.1. Topological Relationships and Their Use

The book chapter Topological Relationships and Their Use highlights that topological relationships are well established for simple 2D objects, while 3D topological operations remain less developed. It identifies the need to support more complex 3D objects, including solids with holes and tunnels, and suggests reusing existing spatial operators as a starting point for extending topological analysis to 3D.

9.5.2. Extension of RCC*-9 to Complex and Three-Dimensional Features and Its Reasoning System

The work described in the paper Extension of RCC*-9 to Complex and Three-Dimensional Features and Its Reasoning System, defines semantics of 3D bodies, regions, and lines, which can support the definition and interpretation of 3D spatial relations and operations. It provides a formal topological foundation that can be interesting for GeoSPARQL 3D.

9.6. Implementations

The related work above primarily address 3D geometry semantics. A complementary aspect is the computational processing of the 3D representations. Implementations of 3D geometry processing provide the algorithms and data structures required to perform operations such as intersection, containment, distance calculation, Boolean operations, and topology analysis.

This section reviews established geometry-processing libraries and systems, including CGAL, SFCGAL, OpenCASCADE, and Topologic, and considers their relevance to the implementation of 3D spatial operations in GeoSPARQL.

9.6.1. CGAL → SFCGAL → PostGIS

CGAL (Computational Geometry Algorithms Library) provides data structures and algorithms for computational geometry, including operations on geometric objects that can serve as a computational foundation for spatial processing. SFCGAL builds on CGAL to provide additional support for 3D and volumetric geometries and operations, including Boolean operations and geometric predicates. SFCGAL is integrated with PostGIS, a spatial database extension for PostgreSQL, allowing such geometry-processing capabilities to be used within a database environment. Together, these technologies illustrate how 3D geometric representations can be connected to computational geometry algorithms and exposed through a spatial database system, providing a relevant implementation perspective for 3D spatial operations in GeoSPARQL.

9.6.2. OpenCASCADE

OpenCASCADE is an open-source software platform for 3D geometric modelling and CAD applications. It provides data structures and algorithms for representing and processing geometric and topological entities, including points, curves, surfaces, solids, and B-Rep models. The platform supports operations such as geometric transformations, Boolean operations, intersections, and shape analysis. Its explicit representation of geometry and topology makes OpenCASCADE relevant for the computational processing of 3D solids and for investigating how B-Rep-based geometry could be supported by a GeoSPARQL 3D implementation.

9.6.2.1. OpenCASCADE-inspired BRep ontology

Perzylo, A., Somani, N., Rickert, M., & Knoll, A. (2015, September). An ontology for CAD data and geometric constraints as a link between product models and semantic robot task descriptions. In 2015 IEEE/RSJ international conference on intelligent robots and systems (IROS) (pp. 4197-4203). IEEE.

https://ieeexplore.ieee.org/abstract/document/7353971

[37]

9.6.2.2. Topologic

Jabi, W., & Chatzivasileiadi, A. (2021, January). Topologic: exploring spatial reasoning through geometry, topology, and semantics. In Formal Methods in Architecture: Proceedings of the 5th International Symposium on Formal Methods in Architecture (5FMA), Lisbon 2020 (pp. 277-285). Cham: Springer International Publishing.

https://topologic.app/

jabi2021topologic

9.6.3. BRep vs mesh/polyhedron

BRep - Curved surfaces - Topology: connected components as shells, solids with inner voids, etc. - Clean APIs due to inheritance: e.g `fn extrude(Topo) -> Topo`, for vertex → edge; edge → face; face → solid; solid → solid - Extra indirections: edge → vertex[] → point - Depending on implementation can be inefficient, e.g outer wire of face not explicitly marked need to be checked wrt infinite point - Data integrity and validation a bit harder

Mesh/polyhedron - Potentially fewer indirections - Triangle meshes robust and well understood - Many different data models though, e.g half-edge (only manifold), indexed faceset (no adjacency info), winged/quad/radial edge

B-Rep and mesh/polyhedron representations differ in how 3D geometry and topology are modelled. B-Rep explicitly represents curved surfaces and topological relationships between vertices, edges, faces, shells, and solids, making it suitable for complex solid models, but it can involve more complex data structures and validation requirements. Mesh and polyhedral representations generally use simpler discrete geometries, often based on triangles, with several possible data structures for representing adjacency and topology, such as half-edge, winged-edge, and indexed face sets. For GeoSPARQL 3D, this distinction is relevant because the underlying representation can affect the precision, robustness, and implementation of spatial operations.

9.7. To consider

https://link.springer.com/article/10.1007/s10845-023-02246-6 Ontology of 3D virtual modeling in digital twin: a review, analysis and thinking

[53]

To help novice engineers understand and scheme 3D virtual modeling in digital twin for future research and applications, this paper reviews 106 digital twin 3D modeling cases with their characteristics, including deployment targets, purposes & roles, collaborative models, data flows, the autonomy of 3D modeling, fidelity, twinning rates, enabling technologies, and enabling tools.

Open standard for particle-mesh data (openPMD)

The openPMD standard, short for open standard for particle-mesh data files is not a file format per se. It provides guidance for meta-data and naming schemes. openPMD provides naming and attribute conventions that allow the exchange of particle and mesh based data from scientific simulations and experiments. Its primary goal is to define a minimal set/kernel of meta information that allows to share and exchange data to achieve portability between various applications and different algorithms, a unified open-access description for scientific

data (publishing and archiving), and a unified description for post-processing, visualization and analysis. If outputs from programs, devices (such as cameras), simulations or post-processed data-sets, contain a minimal set of meta information as provided by openPMD, data can be exchanged between those with minimal effort, with the same tools used for visualization.

[28]

10

REQUIREMENTS FOR GEOSPARQL 3D

10 REQUIREMENTS FOR GEOSPARQL 3D

This section provides an overview of feedback received on the current version of the GeoSPARQL standard (version 1.1) regarding 3D usage. It helps to identify some of the barriers to use, and to outline requirements that have not been addressed that may encourage greater uptake.

10.1. Extension 1: 3D representations

GitHub Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/583

Category

Semantic improvement

Description

GeoSPARQL primarily focuses on spatial representation and querying rather than visualization, formats such as glTF and 3D Tiles could provide a complementary representation and delivery mechanism for the resulting 3D data. GeoSPARQL should include ways to represent 3D data in a knowledge graph. 3D data should be included in common 3D formats and 3D data should be includable as a text literal and a file representation. Where 3D data is intended for web-based visualization, formats should preferably be selected that are well supported by commonly used JavaScript and web-mapping libraries. This would facilitate the visualization and delivery of 3D knowledge-graph data on the web and help establish a practical bridge between BIM and GIS, enabling 3D information originating from BIM to be queried, linked, and visualized together with geospatial data.

Some common formats which could be considered for inclusion are:

- Polygon File Format (PLY)
- Wavefront OBJ Format (OBJ)
- GLTF Format (GLTF)
- X3D Format
- openUSD



Figure 6 — Linked Building Data Server using Semantic Graphs and GLTF-Models

Related work

- X3D Ontology
- GOM
- OMG
- FOG
- CityRDF/OntoCityGML
- ChgOnto

10.2. Extension 2: Topological relations of 3D geometries

GitHub Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/416 https://github.com/opengeospatial/ogc-geosparql/issues/581 https://github.com/opengeospatial/ogc-geosparql/issues/595

Category

Semantic improvement

Description

GeoSPARQL should include ways to represent relations between 3D geometries and relations between 3D geometries and geometries of lower dimensions. The relations should be expressable in property relations and should be queryable using SPARQL extension functions.

— Topological relations in 3D

The topological possibilities in a 3D space differ from those in 2D space. The mathematical formulations used to verify topological relationships in 2D are different from those required in 3D. Furthermore, the algorithms used to analyze topological relationships also differ, as the sequence of analyses needed to identify and evaluate these relationships is not the same in two- and three-dimensional space. As a result, topological analysis in 3D requires a fundamentally different approach from that used in 2D.

The example below illustrates the differences in the possible topological relationships in two- and three-dimensional space for surfaces and lines. All relationships drawn in (2D) exist in both 2D and 3D environments. The relationships illustrated in 3D can only exist in a true 3D environment. It should be noted that not all theoretically possible topological relationships have been assigned a descriptive name. Some relationships are identified only by a numerical identifier within the underlying topological model and therefore cannot be queried directly using named predicates (such as Disjoint, Intersects, or Contains). Investigating which of these unnamed relationships are relevant and should be exposed as explicit query operators in a 3D environment could be an interesting direction for future research.

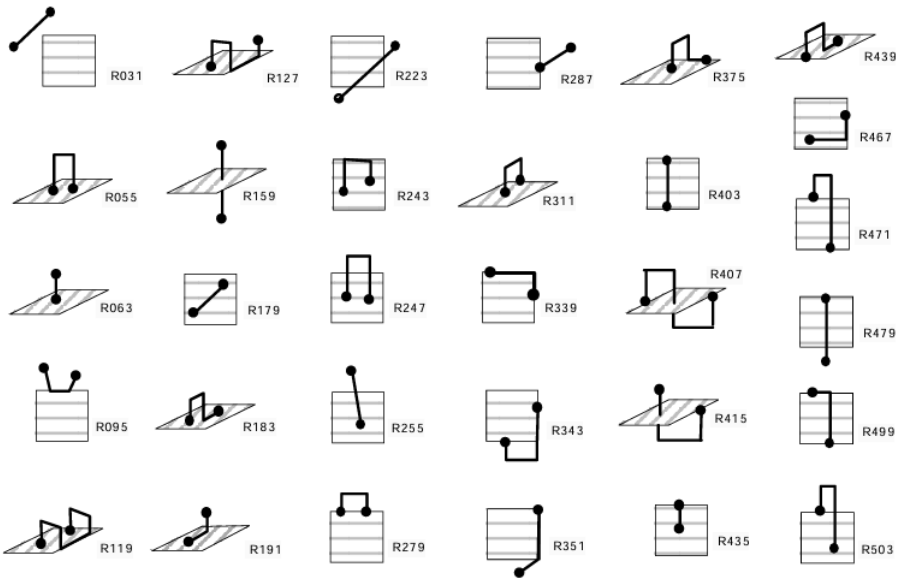


Figure 1: Surface and line in $\mathbb{R}^3$: 31 relationships (19 in $\mathbb{R}^2$, face drawn as reactangle)

Figure 7 — Surface and line topological relations in 2D and 3D space

Disjoint (R031)
Inside (R223)
Touch (R287, R279)
Covers (R439, R435)
Crosses (R159, R243, R247, R255)
Equal (R403, R407, R415)
CoveredBy (R467, R471, R479)
Contains (R179, R183, R191)
Overlaps (R499, R503)
Nameless topological situation (R127, R375, R055, R063, R339, R095, R343, R119, R351)

The semantic definition of a topological relationship, for example disjoint, is identical in both two- and three-dimensional spaces, the outcome of a disjoint function depends on the space in which the geometries are evaluated.

A Query3D evaluates the topological relationship directly in the original three-dimensional Euclidean space. In contrast, a Query2D first projects the geometries onto a two-dimensional plane and subsequently evaluates the topological relationship in this projected space. Consequently, a Query2D does not determine whether two 3D geometries are disjoint in 3D space, but whether their 2D projections are disjoint in 2D space.

As a result, the same pair of geometries may produce different outcomes depending on whether they are evaluated using Query2D or Query3D. Two objects may be disjoint in three-dimensional space while their projections overlap, causing Query3D to return true for Disjoint and Query2D to return false.

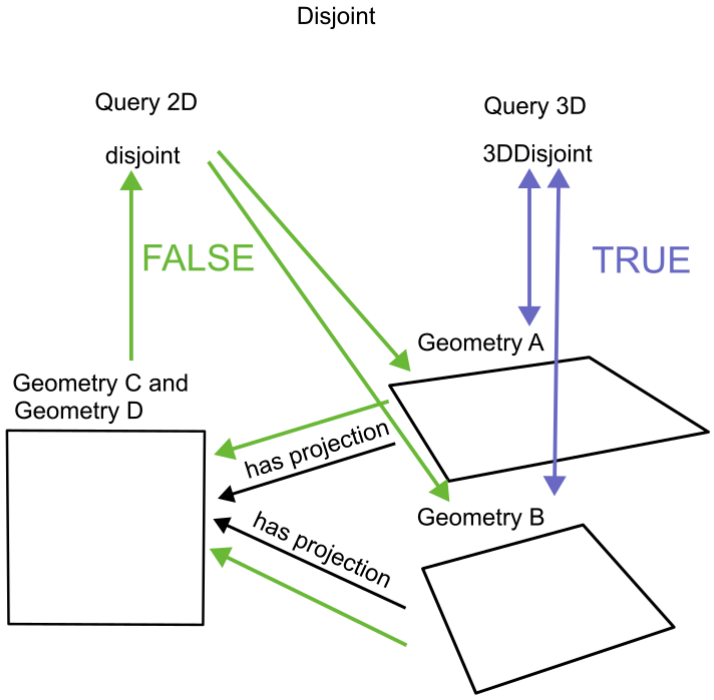


Figure 8 — Disjoint with Query2D and Query3D functions

The example assumes that both geometries are in the same dimension (2D). It is also common to analyze the topological relationship between a geometries of differt dimensions 0D, 1D, 2D or 3D. Such comparisons require both geometries to be represented in the same spatial dimension before a meaningful topological analysis can be performed.

This can be achieved in one of two ways:

— embedding the 2D geometry into 3D space; or

— projecting the 3D geometry into 2D space.

It is recommended that these transformations are performed explicitly and with a consciously selected Z-value.

For example, when ST_Force3D is applied to a 2D geometry, most implementations automatically assign a Z-coordinate of 0, thereby producing a valid 3D geometry. This default value does not necessarily indicate that the geometry is actually located at an elevation of zero. The value merely serves as a placeholder to satisfy the requirements of a three-dimensional coordinate system. After conversion, it is no longer possible to distinguish whether a Z-value of 0 originated from the source data or was automatically introduced during the conversion process. For applications in which elevation is meaningful, this ambiguity may lead to incorrect interpretations or topological analyses. Explicitly specifying the Z-value during the conversion makes this modeling decision transparent and reproducible.

When a Query2D is requested for geometries that exist only as 3D objects without an explicitly defined projection, the system should indicate that the question is not directly applicable. Instead, it should require the user to specify or confirm the projection to be used before performing the topological evaluation. This would make the dimensional c ontext explicit and help prevent misinterpretation of the query results.

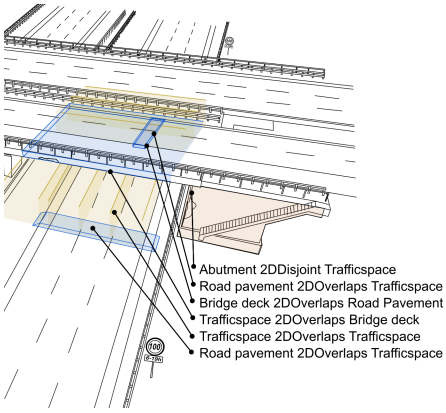


Figure 9 — 2D Topological relations for 3D objects

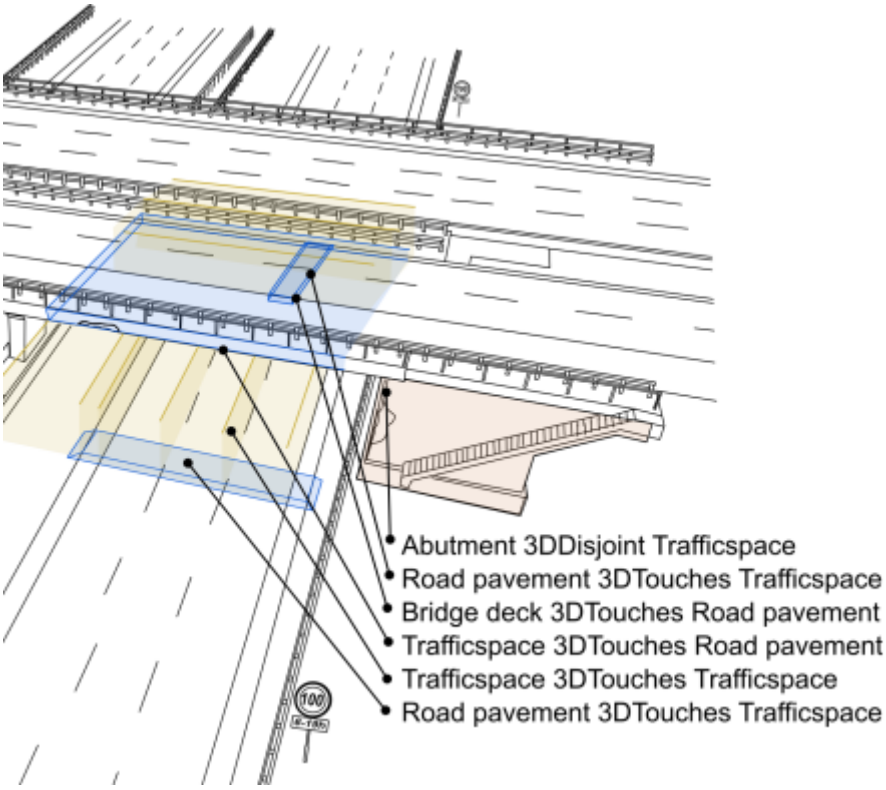


Figure 10 — 3D Topological relations for 3D objects

Related work

- Topological Relationships and Their Use
- Extension of RCC*-9 to Complex and Three-Dimensional Features and Its Reasoning System

10.3. Extension 3: Directional relations of 3D geometries

GitHub Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/684

Category

Semantic improvement

Description

Topological relations describe the relationships between objects independently of perspective. Relative directional relations, on the other hand, describe relationships between objects that

depend on a reference frame or viewpoint. Directional relationships distinguishes different perspectives - viewer-centered, deictic perspective (1 constellation, 2 observers results in 2 spatial relationships) - object-centered, intrisic perspective (object orientation as reference frame) - extrinsic, external perspective (universal reference systems like gravity vector, coordinate systems, axes, linear reference elements)

In 2D, relations such as left of, above, and beside are often sufficient. In 3D, additional relations become relevant, such as above/below and in front of/behind.

Relative directional relations can also provide context in 2D, but many spatial situations can already be described using topology (e.g., inside, touching, overlapping). In 3D environments, however, directional relations become more important because height and depth introduce additional dimensions that cannot be distinguished by topology alone.

For example, describing that a “damage location is underneath a bridge deck” requires both a topological relation (e.g., a 3D contains relation) and a relative directional relation (e.g., below).

This combination of relative location and topological relations has been described by Anne Göbels and Jakob Beetz in the Reloc ontology. Reloc ontology.

This ontology brings natural language closer to the 3D model by formalizing descriptions of relative directions and topological relationships.

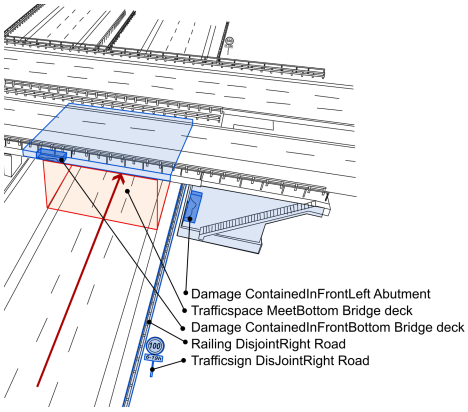


Figure 11 — 3D Topological and Directional relations for 3D objects

Related work

- Reloc Ontology
- SpatialOT Ontology

10.4. Extension 4: Routing

GitHub Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/572

Category

Semantic improvement

Description

GeoSPARQL should be able to query for the shortest route through a 3D space made up of smaller, connected 3D spaces when topological relations are not available. By enabling this, GeoSPARQL can describe spatial data and where features are located, while also supporting analysis/description of how things can move between them. By deriving connections between 3D objects(spaces) using their geometry and openings, GeoSPARQL could support shortest-path and accessibility analysis without requiring all topological relationships to be explicitly modelled. This could be useful for applications such as emergency evacuation, indoor navigation, fire compartmentation and safety analysis, accessibility assessment, and analysing potential pathways for smoke or fire spread.

Ideas: https://isprs-archives.copernicus.org/articles/XLII-4/11/2018/ USING THE COMBINED LADM-INDOORGML MODEL TO SUPPORT BUIILDING EVACUATION

and: 3D pgRouting and Visualization in Cesium JS Using the Integrated Model of LADM and IndoorGML

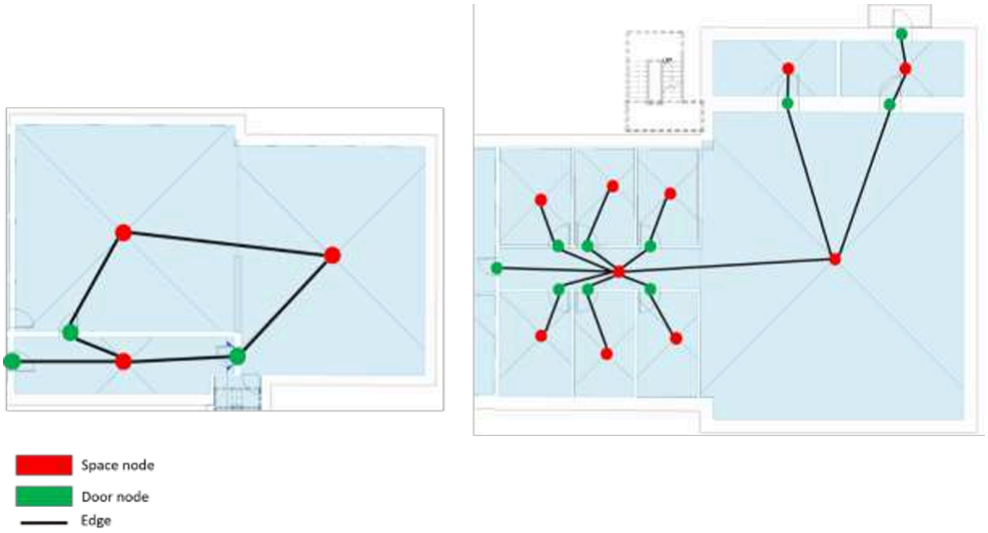


Figure 12 — Space and doors nodes and connecting edges for routing.

Related work

No related work

10.5. Extension 5: Appearance of 3D geometries

GitHub Issue URI

- https://github.com/opengeospatial/ogc-geosparql/issues/584
- https://github.com/opengeospatial/ogc-geosparql/issues/592

Category

Semantic improvement

Description

GeoSPARQL should include ways to represent materials and textures of 3D geometries, so that geometries can be styled accordingly.

Materials include:

- Colors of surfaces with light diffusion parameters
- Images as textures, which are associated with surfaces of the 3D object

The main purpose is not to make GeoSPARQL a visualisation format, but to allow existing rich 3D formats to retain their textures, colours and materials while their geometries can still be queried and analysed spatially. Visualisation is then one possible application of this information. This could also enable queries based on material or texture information, allowing users to combine spatial and material information. For example, a user could query: “Find all features within this area that contain lead.” Visualisation is then one possible application of this information, while material-based spatial analysis provides additional possibilities for applications such as building safety, renovation, and environmental risk assessment.

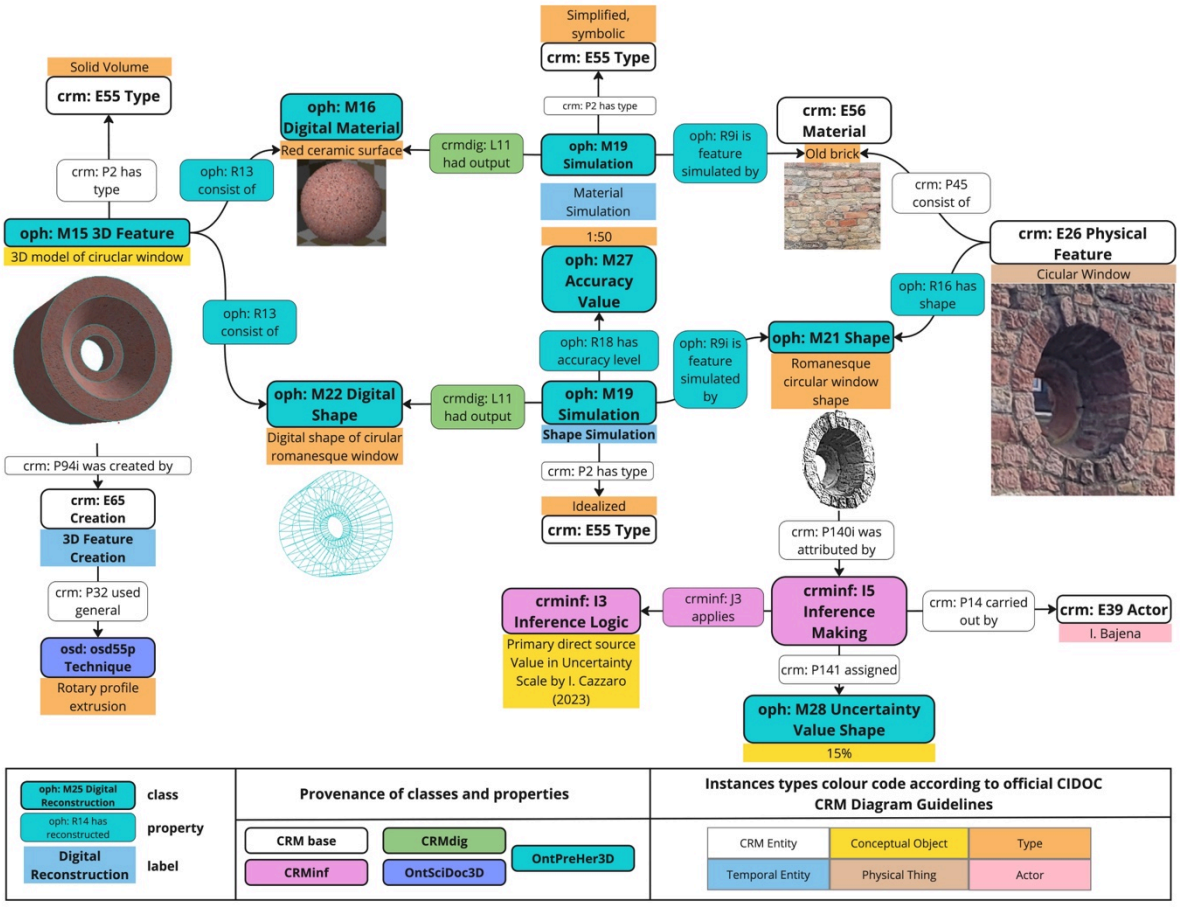


Figure 13 — Diagram illustrating the use of the OntPreHer3D ontology for the hypothetical virtual 3D reconstruction process with the use of a simulation event.

Related work*

X3D IIF 3D OntoPreHer3D

10.6. Extension 6: Multi-component 3D geometries

Github Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/591

Category

Semantic improvement

Description

GeoSPARQL should include ways to define multi-component 3D geometries, whereas each component expresses its own semantics. For example, parts of a building could have different semantics according to the function of the building components and would be classified as such in an RDF graph.

Existing standards such as IFC, CityGML, SML already provide mechanisms for decomposing 3D objects into 3D parts. GeoSPARQL could build on these existing relationships, allowing the geometries of individual components to be queried and analysed together with their semantics. This would enable queries that combine spatial and semantic information, for example, identifying all building components within a specific area that are made of lead, or finding all walls that are part of a particular building and intersect a given spatial region.

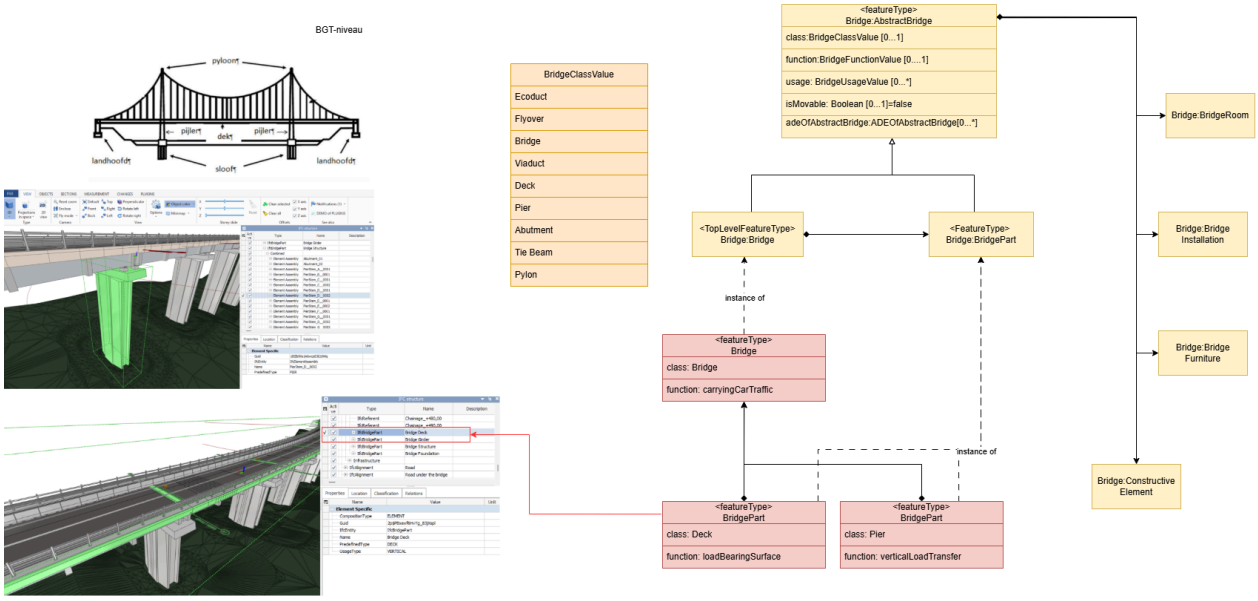


Figure 14 — Model of a bridge where the bridge parts "Pier" and "Deck" contain the geometries of the bridge.

Related work

CityRDF OntoCityGML

10.7. Extension 7: Positioning of 3D geometries

Github Issue URI

- https://github.com/opengeospatial/ogc-geosparql/issues/587
- https://github.com/opengeospatial/ogc-geosparql/issues/588
- https://github.com/opengeospatial/ogc-geosparql/issues/589

— https://github.com/opengeospatial/ogc-geosparql/issues/591

Category

Semantic improvement

Description

GeoSPARQL should include ways to position 3D geometries in a 3D space. Commonly 3D geometries are rotated, translated and scaled using commonly defined operators in computer graphics. Similar operations are needed for the relative positioning of 3D objects in GeoSPARQL, as properties and potentially as functions.

10.8. Extension 8: Alignments of GeoSPARQL 3D

Github Issue URI

— https://github.com/opengeospatial/ogc-geosparql/issues/590

— https://github.com/opengeospatial/ogc-geosparql/issues/574

— https://github.com/opengeospatial/ogc-geosparql/issues/571

Category

Semantic improvement

Description

GeoSPARQL 3D should be aligned to other vocabularies and standard which currently provide 3D support in different knowledge domains. Especially alignments to ifcOWL and the X3D vocabulary would position the GeoSPARQL vocabulary as a link between these different standards.

Related work

10.9. Extension 9: Alignments of Engineering CRS to Geospatial CRS

Github Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/586

Category

Semantic improvement

Description

GeoSPARQL 3D should provide the opportunity to align a local coordinate system in which most 3D geometries are defined with a coordinate reference. While this work might only be partially done within the scope of GeoSPARQL itself, GeoSPARQL should be aligned with the emerging Ontology CRS developments of OGC and provide necessary functions or properties to create the link.

10.10. Extension 10: Geometry Extrusion

Github Issue URI

- https://github.com/opengeospatial/ogc-geosparql/issues/556
- https://github.com/opengeospatial/ogc-geosparql/issues/547

Category

Semantic improvement

Description

GeoSPARQL 3D should provide the opportunity to extrude 2D geometries to 3D geometries and vice versa.

10.11. Extension 11: Geometry Attributes

Github Issue URI

- https://github.com/opengeospatial/ogc-geosparql/issues/568
- https://github.com/opengeospatial/ogc-geosparql/issues/550
- https://github.com/opengeospatial/ogc-geosparql/issues/549
- https://github.com/opengeospatial/ogc-geosparql/issues/548

— https://github.com/opengeospatial/ogc-geosparql/issues/558

Category

Semantic improvement

Description

GeoSPARQL 3D should provide functions and properties that describe essential properties of a 3D Geometry such as its minimum and maximum height, width and depth and its CompactnessRatio.

10.12. Extension 12: Non-topological Query Functions — 3D Extension

Github Issue URI

— https://github.com/opengeospatial/ogc-geosparql/issues/556

Category

Semantic improvement

Description

GeoSPARQL 3D should provide the opportunity to execute non-topological query functions on 2D and 3D geometries commonly used in geospatial databases. Proposed extensions include following functions:

— geometry extrusion to the specified line segment

— geometry extrusion to the specified height

— spatiotemporal geometry extrusion to the specified line segment with specific start and end time

10.13. Extension 13: 3D Geometry available in Open BIM (IFC)

Github Issue URI

https://github.com/opengeospatial/ogc-geosparql/issues/8 https://github.com/opengeospatial/ogc-geosparql/issues/574 https://github.com/opengeospatial/ogc-geosparql/issues/590 https://github.com/opengeospatial/ogc-geosparql/issues/582 https://github.com/opengeospatial/ogc-geosparql/issues/598 https://github.com/opengeospatial/ogc-geosparql/issues/588 https://github.com/opengeospatial/ogc-geosparql/issues/608

Category

Semantic improvement

Description

When using GeoSPARQL with OPEN BIM (IFC), the geometry can either be used directly from the IFC format (natively) or converted to a GeoSPARQL-compatible format. A converter can be used as an intermediate layer, but this introduces several drawbacks: - Additional complexity: an extra component must be maintained. - Performance overhead: geometries need to be extracted and converted. - Synchronization issues: IFC changes require the converted geometry to be regenerated. - Loss of geometric information: IFC can contain richer 3D geometry than formats such as WKT can represent. - Converter dependency: the quality and supported geometry types depend on the conversion tool.

IFC is a standard with a rich set of geometric primitives, with core concepts defined in the geometry, presentation, material, topology and profile resource subschemas. But also outside of these modules, there are various concepts that have geometrical implications such as openings and space boundary relationships in the productextension subschema. The listing Annex C: Overview of IFC Geometry tries to provide a comprehensive overview broken down per the categories below:

- Relational; typically IfcRelationship between elements subtypes, but also clearance geometry which is a potential additional shape representation identifier
- Product-level; resources applicable the IfcProduct such as material-based styling; openings
- Item-level; the individual items that make up representation of the element; which are in turn potentially affected by product-level features such as openings and styling
- Context; provides additional information on individual items such as precision, target scale and semantic use of the geometry
- Geo-referencing; Prevalence is measured as a text-based scan through a set of 1000 files (500 for novel constructs). The number indicates percentage of model featuring this entity; and then the average number of instances for those models that contain any. See second table for more details.

A questionnaire was conducted to identify which types of geometry different stakeholders would like to have available.

10.14. 5.3 Vanilla 3D geometry handling in the Semantic Web

Different ontologies handle geometry in the Semantic Web in different ways. This section describes how 3D geometry is handled by different ontologies, rather than by GeoSPARQL. It compares approaches such as BOT, OMG, FOG, BIMSPARQL, GEOM, OntoBREP, and OntoSTEP, highlighting how each ontology represents, links, or manages geometric information within the Semantic Web.

10.14.1. BOT (Building Topology Ontology)

The Building Topology Ontology distinguishes zones and elements. It also has topological relations as bot:containsZone and bot:hasElement. Any bot:Zone or bot:Element can be assigned a 3D Model using some existing data format for 3D models. Two properties are defined for this. The bot:hasSimple3DModel datatype property links a bot:Zone or bot:Element to a 3D Model encoded as a literal. If the data format is textual, then the lexical form of the 3D Model literal SHOULD be encoded as a Unicode [UNICODE] string, which SHOULD be in Normal Form C [NFC]. If the data format is binary, then the lexical form of the 3D Model literal SHOULD be encoded using a base32 encoding [rfc4648]. The bot:has3DModel object property links a bot:Zone or bot:Element to some IRI that identifies a 3D Model. This 3D Model can then be described using some dedicated RDF vocabulary. Else, the 3D Model IRI could be dereferenceable, and when looking up the IRI one could retrieve a representation of the 3D Model with some existing data format for 3D models.

Figure 15 — Example of a graphical feedback from a request for all wall elements adjacent to a particular space using BOT terminology. The 3D model is described as an OBJ-formatted mesh.

source: https://journals.sagepub.com/doi/pdf/10.3233/SW-200385

10.14.2. OMG (Ontology for Managing Geometry)

The Ontology for Managing Geometry (OMG) is an ontology for attaching geometry descriptions to their corresponding things (e.g. building objects). The OMG is designed to provide three levels of adding the geometry descriptions which can be used and combined flexibly. The levels are designed to enable the attachment of metadata and comprise of the following functionalities:

Level 1: Direct connections between things and their geometry descriptions using either an object property for RDF-based geometry descriptions or a datatype property for non-RDF-

based geometry descriptions resp. links to external files. Level 2: Introduction of an intermediate node between the things and their geometry descriptions. By reifying the geometry description, handling of multiple geometry descriptions of the same things is empowered, metadata (as tags) can be added and dependencies between geometries can be modelled. Level 3: Introduction of an intermediate node between the reified geometry and the geometry description to add geometry states. Thus, the version history of a geometry description can be described and stored.

10.14.3. FOG (File Ontology for Geometry formats)

The File Ontology for Geometry formats (FOG) provides geometry schema specific relations between things (e.g. building objects) and their geometry descriptions. These geometry descriptions can be (1) RDF-based (e.g. using specific ontologies such as GEOM, OntoBREP, etc.), (2) RDF literals containing embedded geometry of existing geometry formats and (3) RDF literals containing a reference to an external geometry file.

such as OBJ, glTF and STEP, to be embedded in RDF or referenced as external files. Thus, while GeoSPARQL primarily describes geospatial geometry, BOT, OMG and FOG provide approaches that are more focused on building and construction geometry and the use of existing geometry formats.

the File Ontology for Geometry formats (FOG), a method for embedding and exchanging different 2D and 3D geometry formats within RDF graphs. FOG allows geometry data to be stored, referenced, and automatically identified, making it easier to exchange building geometry between different software and stakeholders.

10.14.4. LBD Server, IFC2LBD and IFC2LBD-NEO

To make BIM data accessible on the web, the Linked Building Data (LBD) initiative applies Linked Data principles, using RDF, vocabularies, and ontologies to represent building information. To make this information practically accessible, a platform is needed that combines semantic building data with 3D visualisation. The LBD Server addresses this need by allowing users to upload building data, visualise its 3D geometry, and link geometric objects to their underlying semantic information. This enables users to explore both the physical and semantic aspects of a building within a web-based environment.

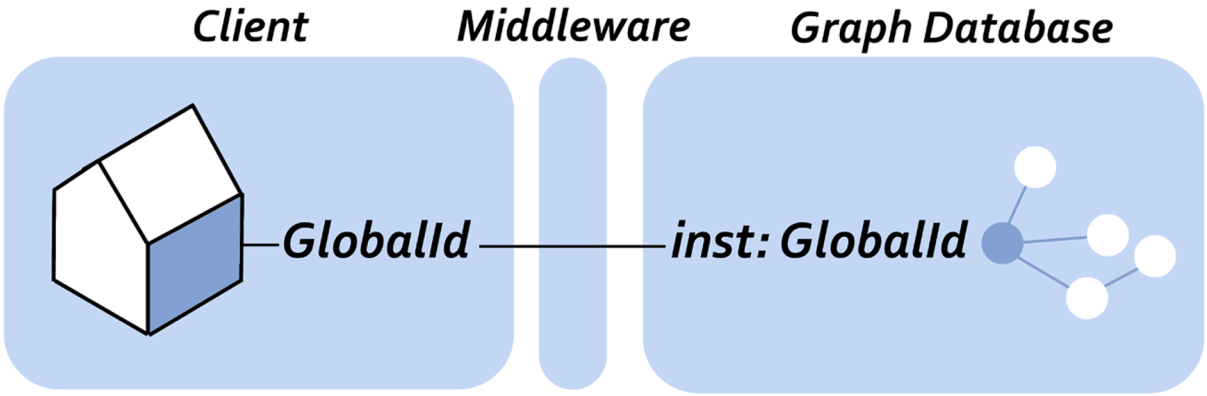


Figure 16 — LBD_Server_connectivity_GLTF_LD.

source: [_3e4260b3605459b8ce220cfb42a3525]

IFC2LBD makes the semantic transformation itself a central component. It converts IFC into RDF using LBD ontologies such as BOT, PRODUCT and PROPS. The result is a graph representation that can be queried and linked on the web. The original IFC-to-LBD work explicitly aimed to produce a simpler, modular representation than directly using the full ifcOWL structure. IFC2LBD-Neo takes this further by redesigning the converter as a modular, staged pipeline. It additionally uses ontologies as BEO, bSDD and OMG/FOG. The geometry is generated separately as glTF, Fragments or Parquet outputs.

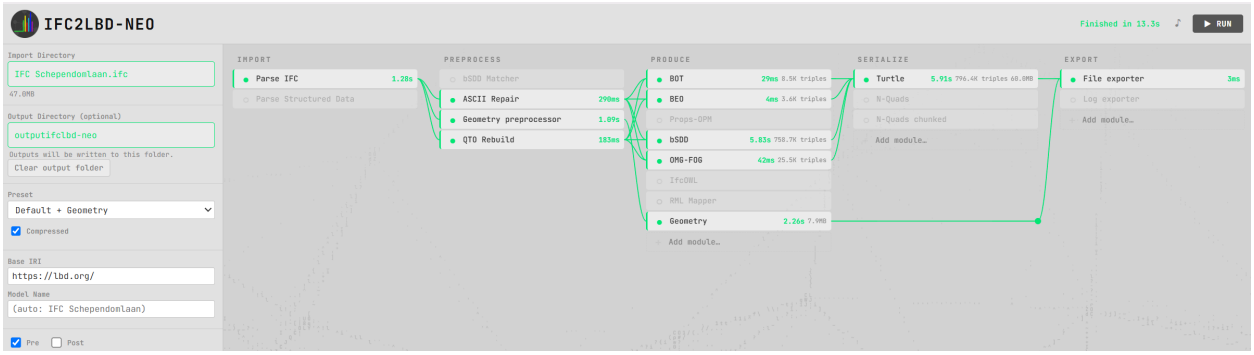


Figure 17 — IFC2LBD Pipeline.

source: https://github.com/ISBE-TUe/IFC2LBD-Neo

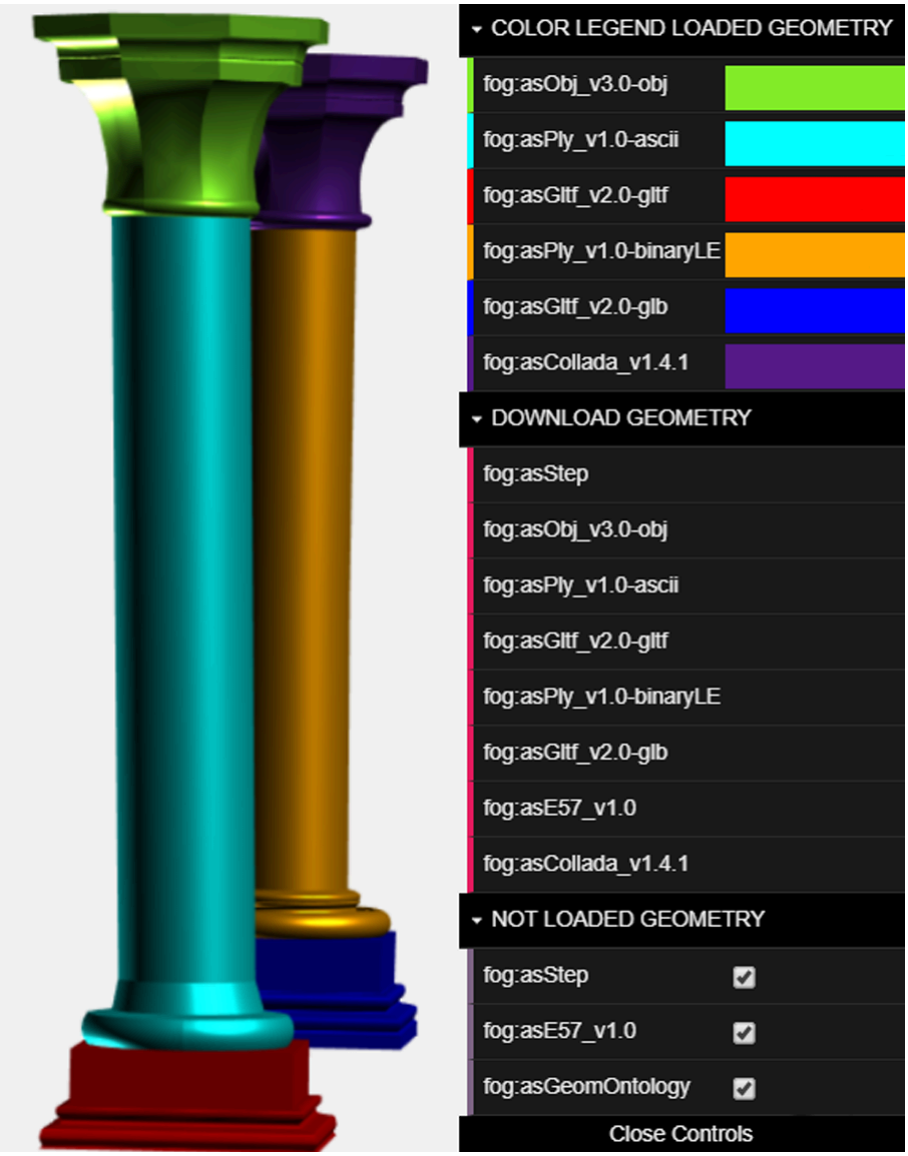


Figure 18 — Demo Application with use of File Ontology for Geometry Formats FOG

source: https://ec-3.org/wp-content/uploads/2025/10/EC32019_166.pdf

10.15. 5.4 Concluding overview of requirements for 3D geometry in the semantic web

The requirements indicate that a GeoSPARQL extension for 3D should provide a common semantic and query layer for heterogeneous 3D representations. Such an extension should support the representation and semantic linking of 3D geometries, 3D topological and directional relations, spatial analysis and measurement, positioning and coordinate reference

systems, visualisation, as well as routing. The standard should interoperate with established geometry formats and domain-specific models. This would allow specialised formats to retain their detailed geometric, visual, and domain-specific capabilities, while GeoSPARQL provides a common framework for representing, linking, discovering, and querying their spatial semantics in the Semantic Web. In this role, GeoSPARQL 3D would function primarily as an interoperability layer between geospatial, BIM, and other 3D domains.

11

BENEFICIARIES AND BENEFITS

11 BENEFICIARIES AND BENEFITS

This section describes the potential beneficiaries and benefits of representing 3D spatial data using semantic and graph technologies. The requirements identified for 3D-GeoSPARQL provide a basis for achieving these benefits by supporting the representation, linking, querying, and reasoning of 3D spatial information across different domains and datasets. Furthermore, a collection of use cases demonstrate how semantic and graph technologies are used together with spatial data to tackle real world problems.

11.1. Semantic-functional value of 3D-GeoSPARQL

Value for data reasoning, formal description, and semantic interoperability

11.1.1. Meaningful creation and querying of 3D geometry

Whether manually or programmatically defining 3D geometries in BIM, GEO or CG environments, the GeoSPARQL standard could provide or connect to vocabulary to represent 3D structures. A 3D cube, for example, can be represented in multiple ways:

- As a set of bounding points, lines, and faces;
- As a base surface with an extrude function;
- As an 3D geometric primitive (e.g., the concept of a 'cube' as a spatial figure).

Once GeoSPARQL offers the means to define and link 3D geometries in these different ways, it will allow both humans and machines to interpret and interact with the data effectively.

11.1.2. Meaningful creation and querying of 3D topological relationships

Is "3D Geometry A" inside, touching, overlapping, intersecting or above "3D Geometry B"?

Being able to describe and use such topological relationships is crucial for conducting spatial analyses, formulating rules, and deriving knowledge from different heterogeneous datasets.

Beyond describing topological relationships, one should be able to query them:

> What is the spatial relationship between 3D Geometry A and 3D Geometry B?

Such queries should return the appropriate topological result (e.g., "intersects"), which is essential for use cases like clash detection in design validation

11.1.3. Derivation of geometric and topological knowledge

From explicitly modeled geometric and topological data, one should be able to infer implicit knowledge.

For instance, if 3D Geometry A is 3D inside 3D Geometry B, then we can infer that:

> A does not “touch” B, and A does not “overlap” with B,

even if these facts are not explicitly stated.

For entailment an application that is compliant with the geo-spatial 3D specification would be able to answer queries like: “Does this tree have a 3D geometry?”

If the tree is defined using a MultiSolid geometry, then the answer should be “yes,” even if this is not explicitly declared as such.

11.2. Value for user experience, visualization, and application use of 3D-GeoSPARQL

11.2.1. Visualization

There is a growing need to visualize how 3D data in BIM and GIS relate to one another. Stakeholders want to see BIM and/or 3D Geo data of a newly planned structure visualized within the 3D Geo and/or BIM context of the existing digital city. A shared vocabulary that can present both domains in an integrated way supports this goal.

11.2.2. Calculation and analysis

Semantic modelled 3D-model enables powerful computation and analysis. Analytical results can be displayed in dashboards operating within the GeoBIM domain. This creates a bridge between asset management (typically GIS-oriented) and project management (typically BIM-oriented), allowing for cross-domain collaboration and decision-making.

11.2.3. Spatial querying

Spatial querying is a key function within a federated GeoBIM Network. Questions such as “What material is located in this area?” or “Where is space available for new cables and pipelines?”

are examples of 3D spatial queries that can be answered when GeoSPARQL 3D is functioning effectively across systems and semantics.

11.2.4. Machine control

A shared semantic GeoBIM Network also supports machine control. The GIS domain typically describes the existing situation, while the BIM domain describes the intended or future situation. This combination can be used to automatically instruct and guide machines in the built environment.

11.2.5. Modeling, simulation and planning

A network of datasets with 3D-data should support the modeling, simulation and planning of future changes to the built environment. Users — whether human or machine — can use the network and GeoSPARQL 3D to propose and model modifications in either BIM or GIS formats. A well-functioning 3D semantic GeoBIM Network enables this kind of forward-looking spatial planning and design.

11.3. Added value of the 3D-GeoSPARQL for Organisations

11.3.1. Brings together different domains (GIS, BIM, CG)

GeoSPARQL 3D can help in bringing together different domains that work with 3D information.

11.3.2. Enables extension to vocabularies from other domains

The vocabulary can have the possiblity to extend to vocabulaires with 3D geometry or topology from other domains, for example the Building Ontology Topology (BOT) or RELOC Ontology. Multi-domain models with georeferenced 3D and 2D city models could be also linked by CityRDF ontology based on CityGML 3.0 standard. GeoSPARQL 3D would be important enabler for data retrieval, update and analysis for such models. In addition to ontologies, it can also establish relationships to 3D file formats from other domains, such as glTF.

11.3.3. Facilitates better interoperability between knowledge domains with spatial components (geography, aerospace, architecture, product design, (bio)chemistry, IoT, ...)

A unified geometric language can help by migrating ideas and methods across fields, accelerating innovation. For example a parametic design algoritm, rule, or llm making use of geometry and topology originated in aerospace, can be re-used in the architecture domain.

12

ANNEX A: ALIGNMENT GEOSPARQL AND OPEN SOURCE RELATIONAL DATABASES WITH GEOSPATIAL SUPPORT .

12 ANNEX A: ALIGNMENT GEOSPARQL AND OPEN SOURCE RELATIONAL DATABASES WITH GEOSPATIAL SUPPORT .

Based on:

GeoSPARQL: PostGIS: https://postgis.net/docs/PostGIS_Special_Functions_Index.html#PostGIS_3D_Functions (PostGIS Functions that support 3D) H2GIS: https://h2gis.readthedocs.io/en/latest/index.html (paragraph Geometry 3D/4D functions) Spatiallite: https://www.gaia-gis.it/gaia-sins/spatialite-sql-latest.html

13

ALIGNMENT: GEOSPARQL AND OPEN SOURCE RELATIONAL DATABASES WITH GEOSPATIAL SUPPORT

13 ALIGNMENT: GEOSPARQL AND OPEN SOURCE RELATIONAL DATABASES WITH GEOSPATIAL SUPPORT

13.1. Formatting functions

Table 4

GeoSPARQL	PostGIS	H2GIS	Spatialite
geof:asGeoJSON	st_asgeojson	ST_AsGeoJson	asgeojson
geof:asGML	st_asgml	st_asgml	asgml
geof:asKML	st_askml	ST_AsKML	askml
geof:asWKT		st_aswkt	aswkt
geof:asEWKB	st_asewkb	AsEWKB	asewkb
	st_ashexewkb		
	ST_AsX3D		asx3d
	link:st_asewkt		asewkt
	st_assvg		assvg
	st_astext	st_astext	astext
			astwkb
	st_asbinary	st_asbinary	asbinary
geof:asDGGS			

geof:asGeocode			
			asfgf
			asgpb
			asencodedpolyline
			atm_astext
			gcp_astext
			proj_asprojstring
			proj_aswkt

13.2. Convert geometries

Table 5

GeoSPARQL	PostGIS	H2GIS	Spatialite
	st_Force2d	st_force2d	
	st_ForceCurve		
	st_force_3d	st_force3d	
	st_force_3dm	st_force3dm	
	st_force_3dz		
	st_force_4d	st_Force4D	
	st_force_collection		
			forceasnull
	st_forcerhr		

CG_ForceLHR		forcelhr
st_forcerSFS		
ForcePolygonCCW		forcepolygonccw
ForcePolygonCW		forcepolygoncw
		normalizelonlat
		project
		spatnetfromgeom
		spatnetfromtgeo
		sridfromauthcrs
st_shift_longitude		st_shift_longitude
st_wkbtosql		st_wkbtosql
st_wkttosql		st_wkttosql
		tinypointencode
		togars
		uncompressgeometry
		xb_getgeometry
	st_projectpoint	
	st_tomultipoint	
	st_tomultisegments	
st_curvetoline		
st_geogfromtext		
st_geogfromwkb		

	st_geographyfromtext		
	st_geometryfromtext		
	st_gmltosql		
	st_linefromwkb		
	st_linestringfromwkb		
	st_linetocurve		

13.3. Create geometries

Table 6

	st_extrude	st_extrude	
	CG_Extrude		
		st_geometryshadow	
geof:boundary	st_boundary	st_boundary	boundary
geof:boundingCircle		st_boundingcircle	
	st_BoundingDiagonal		
	st_minimumboundingcircle		
geof:concaveHull			concavehull
geof:convexHull	st_convexhull	st_convexhull	convexhull
	CG_3DConvexHull		
geof:difference	st_difference	st_difference	difference
	CG_3DDifference		

geof:envelope	st_envelope	st_envelope	envelope
geof:symDifference	st_symdifference	st_symdifference	symdifference
geof:union	st_union	st_union	gunion
	CG_3DUnion		
geof:patchN	st_PatchN		
	CG_MakeSolid		
	ST_LineSubstring		
			atm_create
			atm_createrotate
			atm_createscale
			atm_createtranslate
			atm_createxroll
	st_bdmpolyfromtext		bdmpolyfromtext
			bdmpolyfromwkb
	st_bdpolyfromtext		bdpolyfromtext
			bdpolyfromwkb
			blobfromfile
	st_buildarea		buildarea
			buildcirclembr
			buildmbr
			castautomagic
			casttoblob

			casttodouble
			casttogeometycollection
			casttointeger
			casttolinestring
			casttomulti
		st_tomultiline	casttomultilinestring
			casttomultipoint
			casttomultipolygon
			casttopoint
			casttopolygon
			casttosingle
			casttotext
			casttoxy
			casttoxym
			casttoxyz
			casttoxyzm
	st_collect	st_collect	collect
	st_collectionextract	st_collectionextract	collectionextract
			createnetwork
			createroutingnodes
			createtopogeo
			createtopology

			dissolvepoints
			dissolvesegments
			elementarygeometries
			extractmultilinestring
			extractmultipoint
			extractmultipolygon
			geomcollfromtext
			geomcollfromwkb
			geometrycollectionfromtext
			geometrycollectionfromwkb
			geometrypointencode
	st_geomfromewkb		geomfromewkb
	st_geomfromewkt		geomfromewkt
			geomfromexifgpsblob
			geomfromfgf
	st_GeomFromGeoJSON		geomfromgeojson
	st_geomfromgml	st_geomfromgml	geomfromgml
			geomfromgpb
	st_geomfromkml		geomfromkml
	st_geomfromtext	st_geomfromtext	geomfromtext
	st_geomfromwkb	st_geomfromwkb	geomfromwkb
			geosconcavehull

			gpkgmakepoint
			gpkgmakepointm
			gpkgmakepointz
			gpkgmakepointzm
			gunion
			hexagonalgrid
			hilbertcode
			linefromencodedpolyline
	st_linefromtext	st_linefromtext	“linefromtext linestringfromtext”
			“linefromwkb linestringfromwkb”
			linesfromrings
			linestringfromtext
			linestringfromwkb
			makearc
			makecircle
			makecircularsector
			makecircularstripe
		st_makeellipse	makeellipse
			makeellipticarc
			makeellipticsector
	st_makeline	st_makeline	makeline
	st_makepoint	st_makepoint	makepoint

	st_makepointm		makepointm
			makepointz
			makepointzm
	st_makepolygon	st_makepolygon	makepolygon
			mlinefromtext
			mlinefromwkb
			mpointfromtext
	st_mpointfromtext	st_mpointfromtext	“mpointfromtext multipointfromtext”
			“mpointfromwkb multipointfromwkb”
			mpolyfromtext
			mpolyfromwkb
			multilinestringfromtext
			multilinestringfromwkb
			multipolygonfromtext
			multipolygonfromwkb
			orientedenvelope
	st_pointfromtext	st_pointfromtext	pointfromtext
	st_pointfromwkb	st_pointfromwkb	pointfromwkb
	st_pointonsurface	st_pointonsurface	pointonsurface
			polyfromtext
			polyfromwkb
			polygonfromtext

			polygonfromwkb
	st_polygonize	st_polygonize	polygonize
			squaregrid
	st_point	st_point	st_point
			topogeo_polygonize
			triangulargrid
			voronojdiagram
		st_isovist	
		st_linefromwkb	
		st_makeenvelope	
		st_makegrid	
		st_makegridpoints	
		st_minimumrectangle	
	st_mlinefromtext	st_mlinefromtext	
	st_mpolyfromtext	st_mpolyfromtext	
		st_octogonalenvelope	
	st_polygonfromtext	st_polyfromtext	
		st_polyfromwkb	
		st_ringbuffer	
		st_ringsidebuffer	
		st_sidebuffer	
		st_tessellate	

	CG_Tesselate		
	st_linefrommultipoint		
	st_makebox2d		
	st_makebox3d		
	st_makeenvelope		
	st_multi		
	st_polygon		

13.4. Edit geometries

Table 7

GeoSPARQL	PostGIS	H2GIS	Spatialite
		st_addz	
	AddGeometryColumn		addgeometrycolumn
	st_addmeasure		addmeasure
	st_addpoint	st_addpoint	addpoint
			addedgemodface
			addedgenewfaces
			addfdogeometrycolumn
	DropGeometryColumn		
			addisoedge
			addisonetnode

			addisonode
			addlink
			addtemporarygeometrycolumn
			xb_addfileid
			xb_addparentid
		st_interpolate3dline	
	ST_InterpolatePoint		interpolatepoint
	ST_LineInterpolatePoint		
	ST_LineInterpolatePoints		
	ST_3DLineInterpolate Point		
			line_interpolate_equidistant_ points
	st_line_interpolate_point		line_interpolate_point
			trajectoryinterpolatepoint
		st_multiplyz	
	st_reverse	st_reverse	reverse
		st_reverse3dline	
	UpdateGeometrySRID		
			storedvar_updatetitle
			storedvar_updatevalue
			se_updatevectorcoverageextent
			storedproc_updatesqlbody
			storedproc_updatetitle

			topogeo_updateseeds
			toponet_updateseeds
			updatelayerstatistics
			updatemetacatalogstatistics
		st_zupdatelineextremities	
		st_updatez	
geof:buffer	st_buffer	st_buffer	buffer
geof:transform	st_transform	st_transform	transform
geof:metricBuffer			
geof:transform CRS84			
	CG_ChaikinWmooting		
	ST_Scroll		
	ST_WrapX		
			atm_invert
			atm_multiply
			atm_rotate
			atm_scale
			atm_transform
			atm_translate
			atm_xroll
			atm_yroll
			blobtofile

			bufferoptions_reset
			bufferoptions_setendcapstyle
			bufferoptions_setjoinstyle
			bufferoptions_setmitrelimit
			bufferoptions_ setquadrantsegments
			changeedgegeom
			changelinkgeom
			compressgeometry
			discardfdogeometrycolumn
			discardgeometrycolumn
			drapeline
			drapelineexceptions
			dropnetwork
			droptopology
			dropvirtualgeometry
			ensureclosedrings
	st_expand	st_expand	expand
			gcp_transform
			gcp2atm
			geosdensify
			geosmakevalid
			gpkginsertepsgsrid

			inittopogeo
			inittoponet
	st_line_substring		line_substring
	st_linemerge	st_linemerge	linemerge
			linescutatnodes
			loginetfromtgeo
	st_makevalid	st_makevalid	makevalid
			makevaliddiscarded
			modedgeheal
			modedgesplit
			modgeolinksplit
			modlinkheal
			modloglinksplit
			moveisonetnode
			moveisonode
			newedgeheal
			newedgessplit
			newgeolinksplit
			newlinkheal
			newloglinksplit
			offestcurve
			reduceprecision

	reflectcoordinates
	reflectcoords
	remedgemodface
	remedgenewface
	remisoedge
	remisonetnode
	remisonode
	removelink
st_removepoint	removepoint
	ringscutatnodes
	rotatecoordinates
	rotatecoords
	sanitizegeometry
	scalecoordinates
	scalecoords
st_segmentize	segmentize
	setdecimalprecision
	setendpoint
	setmultiplepoints
st_setpoint	setpoint
	setstartpoint
	shiftcoordinates

			shiftcoords
	st_simplify	st_simplify	simplify
	CG_Simplify		
	st_ simplifypreservetopology	st_simplifypreservetopology	simplifypreservetopology
			singlesidedbuffer
		st_snap	snap
			snapandsplit
	st_snaptogrid		snaptogrid
		st_split	split
			splitleft
			splitright
			st_cutter
	st_node	st_node	st_node
			st_subdivide
	st_translate	st_translate	st_translate
	CG_3DTranslate		
			swapcoordinates
	swapordinates		
			swapcoords
			transformxy
			transformxyz
	st_UnaryUnion		unaryunion

		st_densify
		st_explode
	ST_FlipCoordinates	st_flipcoordinates
		st_normalize
		st_precisionreducer
		st_ removeduplicatedcoordinates
		st_removeholes
		st_removepoints
	ST_RemoveRepeated Points	st_removerepeatedpoints
	st_rotate	st_rotate
	CG_3DRotate	
	st_rotatex	
	CG_RotateX	
	CG_RotateY	
	CG_RotateZ	
	st_rotatey	
	st_rotatez	
	st_scale	st_scale
	CG_3DScale	
	CG_3DScaleAroundCenter	
		st_split
	st_affine	

	st_dump
	st_DumpPoints
	st_dumprings
	st_dumpsegments
	st_memunion
	st_transscale

13.5. Topological Relations

Table 8

GeoSPARQL	PostGIS	H2GIS	Spatialite
geof:sfContains	st_contains	st_contains	contains
geof:ehCoveredBy	st_coveredby		coveredby
geof:ehCovers	st_covers	st_covers	covers
geof:sfCrosses	st_crosses	st_crosses	crosses
geof:sfDisjoint	st_disjoint	st_disjoint	mbrdisjoint
geof:sfEquals	st_equals	st_equals	equals
	TG_Equals		
geof:intersection	st_intersection	st_intersection	intersection
	CG_3DIntersection		
geof:sfIntersects	st_intersects	st_intersects	intersects
	tg_intersects		

	ST_3DIntersects		
geof:sfOverlaps	st_overlaps	st_overlaps	overlaps
	&&&		
	&&&(geometry,gidx)		
	&&&(gidx, geometry)		
	&&&(gidx,gidx)		
geof:sfTouches	st_touches	st_touches	touches
geof:sfWithin	st_within	st_within	within
	st_dfullywithin		
	st_3DDFullyWithin		
geof:ehContains			
geof:ehDisjoint			
geof:ehEquals			
geof:ehInside			
geof:ehMeet			
geof:ehOverlap			
geof:rcc8dc			
geof:rcc8ec			
geof:rcc8eq			
geof:rcc8ntpp			
geof:rcc8ntppi			
geof:rcc8po			

geof:rcc8tpp			
geof:rcc8tppi			
geof:relate	st_relate	st_relate	relate
	st_CPAWithin		
			buildmbrfilter
	st_closestpoint	st_closestpoint	closestpoint
	ST_ClosestPointOfApproach		
	ST_3DClosestPoint		
		st_envelopesintersect	envelopesintersects
			filtermbrcontains
			filtermbrintersects
			filtermbrwithin
			garsmbr
			getfacebypoint
			getfaceedges
			getfacegeometry
			getlinkbypoint
			getnetnodebypoint
			getnodebypoint
			ptdistwithin
			relatematch
	st_setsrid	st_setsrid	setsrid

			sharedpaths
			validatetopogeo
			validlogicalnet
			validspatialnet
	st_dwithin	st_dwithin	
		st_lineintersector	
		st_orderingequals	
	@		
	&&		
	&<		
	&<|		
	&>		
	<<		
	<<|		
	=		
	>>		
	|&>		
	|>>		
	~		
	~=		
	st_containsproperly		
	st_linecrossingdirection		

	st_orderingequals		
	st_point_inside_circle		

13.6. Measure

Table 9

GeoSPARQL	PostGIS	H2GIS	Spatialite
geof:area	st_area	st_area	area
	CG_3DArea	st_3darea	
geof:perimeter	st_perimeter	st_perimeter	perimeter
	st_perimeter2d		
	st_perimeter3d	st_3dperimeter	
geof:metricPerimeter			
geof:length	st_length	st_length	length
	ST_3DLength	ST_3DLength	3dlength
geof:metricArea			
geof:metricLength			
			geodesic length
			geodesicarcarea
			geodesicarcheight
			geodesicarclength
			geodesiccentralangle

			geodesicchordlength
			great circle length
			linestringavgsegmentlength
	st_length_spheroid		
	st_length2d		
	st_length2d_spheroid		
	st_length3d_spheroid		
			linestringminsegmentlength
		st_svf	
		st_sunposition	
geof:distance	st_distance	st_distance	distance
	st_distanceCPA		
geof:metricDistance			
	ST_3DDistance		3ddistance
	CG_Volume		
	ST_GeometricMedian		
			abs
			acos
			asin
			atan
			atan2
	st_azimuth	st_azimuth	azimuth

			ceil
			ceiling
			centimeter
			circularity
			cos
			cot
			curvosityindex
			dd to dms
			decimeter
			degrees
			distancewithin
			dms to dd
			downhillheight
			exp
	st_extent	st_extent	extent
	ST_3DExtent		
			floor
			frechetdistance
			gcp_compute
			geosmaximuminscribedcircle
			geosminimumboundingcenter
			geosminimumboundingcircle

			geosminimumboundingradius
			geosminimumclearance
			geosminimumclearanceline
			geosminimumrotatedrectangle
			geosminimumwidth
			getlayerextent
			getvirtualtableextent
	st_hausdorffdistance		hausdorffdistance
	st_line_locate_point		line_locate_point
	st_locate_along_measure		locatealongmeasure
	st_locate_between_measures		locatebetweenmeasures
	st_maxdistance	st_maxdistance	maxdistance
	3DMaxDistance		3dmaxdistance
	st_shortestline	st_shortestline	shortestline
	st_3dshortestline		
			updownheight
			uphillheight
		st_closestcoordinate	
		st_furthestcoordinate	
		st_locatealong	
	st_longestline	st_longestline	
	st_3DLongestLine		

	st_distance_sphere		
	st_distance_spheroid		

13.7. Count

Table 10

GeoSPARQL	PostGIS	H2GIS	Spatialite
geof:numGeometries	st_numgeometries	st_numgeometries	numgeometries
geof:numPoints	st_numpoints	st_numpoints	numpoints
	st_numpatches		
geof:numInteriorRing			
geof:numPatches			
	ST_NumCurves		
	st_npoints	st_npoints	npoints
	st_nrings		nrings
	st_numinteriorring	st_numinteriorring	numinteriorring
	st_numinteriorrings	st_numinteriorrings	numinteriorrings
	st_locatebetweenelevations		

13.8. Properties

Table 11

GeoSPARQL	PostGIS	H2GIS	Spatialite
		st_is3d	is3d
geof:x	st_x	st_x	x
geof:maxX	st_xmax	st_xmax	
geof:minX	st_xmin	st_xmin	
geof:y	st_y	st_y	y
geof:maxY	st_ymax	st_ymax	
geof:minY	st_ymin	st_ymin	
geof:z	st_z	st_z	z
geof:maxZ	st_zmax	st_zmax	maxz
geof:minZ	st_zmin	st_zmin	minz
geof:m	st_m	ST_M	m
geof:centroid	st_centroid	st_centroid	centroid
geof:geometryN	st_geometryn	st_geometryn	geometryn
geof:getSRID	st_srid	st_srid	srid
	geof:isEmpty	st_isempty	st_isempty
isempty	geof:isSimple	st_issimple	st_issimple
issimple	geof:endPoint	st_endpoint	st_endpoint
endpoint	geof:exteriorRing	st_exteriorring	st_exteriorring
exteriorring	geof:interiorRingN	st_interiorringn	st_interiorringn
interiorringn	geof:isRing	st_isring	st_isring
isring	geof:pointN	st_pointn	st_pointn

pointn	geof:startPoint	st_startpoint	st_startpoint
startpoint	geof:easting		
	geof:northing		
	geof:isClosed	st_isclosed	st_isclosed
isclosed		CG_IsPlanar	
		CG_IsSolid	
		ST_IsCollection	
atm_determinant			
atm_isinvertible			
atm_isvalid			
bufferoptions_getendcapstyle			
bufferoptions_getjoinstyle			
bufferoptions_getmitrelimit			
bufferoptions_getquadrantsegments			
checkwithoutrowid			
checkspatialindex			
checkspatialmetadata		st_coorddim	st_coorddim
coorddimension		st_dimension	st_dimension
dimension			
gcp_isvalid		st_geohash	
geohash		Geometrytype	

geometrytype
getdecimalprecision
getgpkgamphibiousmode
getgpkgmode
getisometadataid
getpointindex
gpkg_isassignable
gpkggetimagetype
gpkggetnormalrow
gpkggetnormalzoom
hasepsg
hasfreexl
hasgcp
hasgeocallbacks
hasgeopackage
hasgeos
hasgeos3100
hasgeos3110
hasgeosadvanced
hasgeosonlyreentrant
hasgeosreentrant
hasgeostrunk

hasiconv		
hasknn		
haslibxml2		
hasmathsql		
hasminizip		
hasproj		
hasproj6		
hasrouting		
hasrttopo		
hastopology		
isgeometryblob		
iscompressedgeometryblob		
ismeasured	ST_IsPolygonCCW	
ispolygonccw	ST_IsPolygonCW	
ispolygoncw	st_isvalid	st_isvalid
isvalid	st_IsValidTrajectory	
		st_isvaliddetail
isvaliddetail	st_isvalidreason	st_isvalidreason
isvalidreason		
isvalidtrajectory		
maxm		
mbrmaxx		

mbrmaxy		
mbrminx		
mbrminy		
minm	st_ndims	
ndims		
selfintersections		st_compactnessratio
	st_geometrytype	st_geometrytype
		st_geometrytypecode
		st_holes
		st_isrectangle
	box3d	
	st_curveN	
	st_estimated_extent	
	st_extent3d	
	st_hasarc	
	st_hasM	
	st_hasZ	
	st_summary	
	st_zmflag	

13.9. Topography

Table 12

GeoSPARQL	PostGIS	H2GIS	Spatialite
		st_drape	
		st_triangleaspect	
		st_trianglecontouring	
		st_triangledirection	
		st_triangleslope	
		st_voronoi	
	CG_ApproximateMedialAxis		
	CG_Orientation		
	CG_StraightSkeleton		
	CG_Visibility		
			geoslargestemptycircle
	st_mem_size		

13.10. Triangulation

Table 13

GEOSPARQL	POSTGIS	H2GIS	SPATIALITE
		st_constraineddelaunay	constraineddelaunaytriangulation
	st_DelaunayTriangles		
	CG_ConstrainedDelaunayTriangles		

GEOSPARQL	POSTGIS	H2GIS	SPATIALITE
			delaunaytriangulation
		st_delaunay	
	CG_AlphaWrapping		

13.11. Export/import

Table 14

GeoSPARQL	PostGIS	H2GIS	Spatialite
			exportdbf
			exportdxf
			exportgeojson2
			exportkml
			exportshp
			importdbf
			importdxf
			importdxffromdir
			importgeojson
			importshp
			importwfs
			importxls
			importzipdbf

				importzipshp
				initfdospatialmetadata
				initspatialmetadata
				initspatialmetadatafull
				insertepsgsrid
				topogeo_exporttopolayer
				topogeo_fromgeotable
				topogeo_fromgeotableext
				topogeo_fromgeotablenoface
				topogeo_fromgeotablenofaceext
				topogeo_togeotable
				topogeo_togeotablegeneralize
				toponet_fromgeotable
				toponet_togeotable
				toponet_togeotablegeneralize
	find_srid			

13.12. Other

Table 15

GeoSPARQL	PostGIS	H2GIS	Spatialite
			autofdostart

	autofdostop
	autogpkgstart
	autogpkgstop
	check_strict_sql_quoting
	checkduplicaterows
	checkgeopackagemetadata
	checkshadowedrowid
	clonetable
	countunsafetriggers
	createclonedtable
	createisometadatatables
	creatembrcache
	createmetacatalogtables
	createmissingrasterlite2columns
	createmissingsystemtables
	createrastercoveragestable
	createrouting
	createrouting_getlasterror
	createspatialindex
	createstylingtables
	createtemporaryspatialindex
	createtopotables

	createuuid
	createvectorcoveragestables
	decodeurl
	dirnamefrompath
	disablegpkgamphibiousmode
	disablegpkgmode
	disablepause
	disablespatialindex
	disabletinypoint
	droptable
	enablegpkgamphibiousmode
	enablegpkgmode
	enablepause
	enabletinypoint
	encodeurl
	eval
	fileextfrompath
	filenamefrompath
	freexl_version
	fullfilenamefrompat
	geos_version
	getcuttermessage

	getdbobjectscope
	getlastnetworkexception
	getlasttopologyexception
	getmimetype
	gpkgcreatebasetables
	gpkgcreatetilestable
	gpkgcreatetileszoomlevel
	indian chain
	indian foot
	indian yard
	international chain
	international fathom
	international foot
	international inch
	international link
	international nautical mile
	international statute mile
	international yard
	invalidatelayerstatistics
	isdecimalnumber
	isexifblob
	isexifgpsblob

	isgifblob
	isinteger
	isjp2blob
	isjpegblob
	islowascii
	isnumber
	ispauseenabled
	ispdfblob
	ispngblob
	istiffblob
	istinypointblob
	istinypointenabled
	isvalidgpb
	iswebpblob
	iszipblob
	kilometer
	libxml2_version
	ln
	log
	log10
	log2
	makestringlist

	md5checksum
	md5totalchecksum
	millimeter
	pause
	pi
	pow
	power
	proj_getdatabasepath
	proj_guesssridfromshp
	proj_guesssridfromwkt
	proj_guesssridfromzipshp
	proj_setdatabasepath
	proj_version
	radians
	rebuildgeometrytriggers
	recoverfdogeometrycolumn
	recovergeometrycolumn
	recoverspatialindex
	recreateisometarefstriggers
	recreaterastercoveragestriggers
	recreatestylingtriggers
	recreatetopotriggers

	recreatevectorcoveragestriggers
	registerdatalicense
	registerisometadata
	registervirtualgeometry
	removeduplicaterows
	removeextraspaces
	renamecolumn
	renamedatalicense
	renametable
	rl2_mapconfigurationabstractn
	rl2_mapconfigurationnamen
	rl2_mapconfigurationtitlen
	rl2_nummapconfigurations
	rl2_registermapconfiguration
	rl2_reloadmapconfiguration
	rl2_unregistermapconfiguration
	rttopo_version
	sequence_currval
	sequence_lastval
	sequence_nextval
	sequence_setval
	setdatalicenseurl

		sign
		sin
		spatialite_target_cpu
		spatialite_version
		sqrt
		stddev_pop
		stddev_samp
		tan
		u.s. chain
		u.s. foot
		u.s. inch
		u.s. statute mile
		u.s. yard
		unregisterdatalicense
		upgradegeometrytriggers
		var_pop
		var_samp
st_accum	st_accum	
	st_googlemaplink	
	st_osmmaplink	

14

ANNEX B: ALLIGNMENT OF GEOMETRY

14 ANNEX B: ALLIGNMENT OF GEOMETRY

Overview of ISO 19107 GML IFC STEP OpenCascade WKT X3D Geometries

Dit overzicht van geometriën laat zien welke geometrieen dicht bij elkaar zitten. Het is geen 10% match

Points:

Table 16

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
		IfcCartesianPoint	cartesian_point	Geom2d_Cartesian Point Geom_Cartesian Point	
GM_ Point	gml: Point			Geom2d_Point Geom_Point	POINT (x y (z) (m))
			cylindrical_point		
			spherical_point		
			polar_point		
		IfcPointByDistance Expression			
		IfcPointOnCurve	point_on_curve		
		IfcPointOnSurface	point_on_surface		
			point_in_volume		
			point_replica		
			degenerate_pcurve		
			evaluated_degenerate_ pcurve		

Lines and Curves:

Table 17

ISO 19107	GML	IFC	STE	OpenCascade	WKT
GM_Curve	gml:Curve				
GM_Orientable Curve	gml:Orientable Curve				
GM_Arc	gml:Arc				CIRCULARSTRING (x y (z)(m))
GM_Arc ByBulge	gml:ArcByBulge				
GM_ArcString	gml:ArcString				
GM_ArcString ByBulge	gml:ArcString ByBulge				
	gml:Circle ByCenterPoint				
	gml:CubicSpline				
	gml:Knot				
	gml:Affine Placement				
	gml:Abstract Surface				
	gml:Polygon				POLYGON(x y (z) (m))
GM_Polygon	gml:Polygonpatch				
GM_Ring	gml:Ring				
	gml:AbstractRing				
	gml:LinearRing				
	gml:Surface				
	gml:Patches				
	gml:AbstractRing				
GM_Orientable Surface	gml:Orientable Surface				

	gml:Bspline	IfcBSplineCurveWithKnots	B_ splir curv with knot	Geom2d_BSpline Curve Geom_BSpline Curve
		IfcRationalBSplineCurveWith Knots		
GM_Bezier	gml:Bezier		bezi curv	Geom2D_Bezier Curve Geom_Bezier Curve
			ratic b_ splir curv	
			loca b_ splir	
			unif curv	
			quas unif curv	
			loca refir splir curv	
			ratic loca refir splir curv	
			trim curv	Geom2d_Trimmed Curve Geom_Trimmed Curve
		IfcCompositeCurveOnSurface		
		IfcBoundaryCurve	bou curv	
		IfcOuterBoundaryCurve	oute bou curv	

		IfcGradientCurve			
		IfcSegmentedReferenceCurve			
		IfcIndexedPolyCurve			
GM_LineString	gml:LineString Segment				LINESTRING(x y (z)(m))
	gml:LineString	IfcPolyline	poly	TopoDS_Wire	
		IfcTrimmedCurve			
GM_Circle	gml:Circle	IfcCircle	circl	Geom2d_Circle Geom_Circle	
		IfcElipse	elips	Geom2d_Ellipse Geom_Ellipse	
		IfcLine	line	Geom2d_Line Geom_Line	
			hype	Geom2d_ Hyperbola Geom_Hyperbola	
			para	Geom2d_Parabola Geom_Parabola	
	gml:OffsetCurve	IfcOffsetCurve2D	offs curv 2d	Geom2d_Offset Curve Geom_Offset Curve	
		IfcOffsetCurve3D	offs curv 3d		
		IfcOffsetCurveByDistances			
		IfcPcurve	pcur		
			bou pcur		
		IfcPolynomialCurve			
		IfcClothoid	clotl		
			circu invo		

	IfcCosineSpiral			
	IfcSecondOrderPolynomialSpiral			
	IfcThirdOrderPolynomialSpiral			
	IfcSeventhOrderPolynomialSpiral			
	IfcSineSpiral			
	IfcSurfaceCurve	surf curv		
	ifcIntersectionCurve	inte curv		
		sear curv		
		bou surf curv		
		com curv on_ surf		
		curv repli		
	IfcSeamCurve			
			Geom2d_Vector WithMagnitude Geom_Vector WithMagnitude	
				shapeExtend_ ComplexCurve

Surface:

Table 18

ISO 19107	GML	IFC	OpenCascade	WKT
GM_Surface	gml:Surface			
GM_Polyhedral Surface	gml:PolyhedralSurface			

GM_Triangulated Surface	gml:TriangulatedSurface	IfcTriangulatedFaceSet	
GM_Tin	gml:Tin		
GM_SurfacePatch	gml:AbstractSurfacePatch		
	gml:Triangle		
	gml:Rectangle		
GM_PointGrid	gml:PointGrid		
GM_Parametric CurveSurface	gml:AbstractParametricCurve Surface		
GM_GriddedSurface	gml:AbstractGriddedSurface		
		IfcPlane	Geom_Plane
		IfcCylindricalSurface	Geom_Cylindrical Surface
			Geom_ConicalSurface
		IfcSphericalSurface	Geom_Spherical Surface
		IfcToroidalSurface	Geom_ToroidalSurface
		IfcBSplineSurface	Geom_BSplineSurface
		IfcBSplineSurfaceWithKnots	

		Geom_BezierSurface
	IfcCurveBoundedPlane IfcCurveBoundedSurface	
		Geom_OffsetSurface
		Geom_Surface
		Geom_Osculating Surface
	IfcRectangularTrimmed Surface	Geom_Rectangular TrimmedSurface
	IfcSectionedSurface	

Primitives:

A primitive (like a block or ellipsoid) can be defined with just a few parameters. It is more efficient than describing the same shape with triangles (mesh) or surfaces (B-rep). It keeps files smaller and mathematically exact.

Table 19

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
		IfcBlock	block	TopoDS_Solid (BRepPrimAPI_MakeBox)	
		IfcRectangularPyramid	rectangular_pyramid		
GM_Cone	gml:Cone	IfcRightCircularCone	right_circular_cone eccentric_cone	TopoDS_Solid (BRepPrimAPI_MakeCone)	
GM_Cylinder	gml:Cylinder	IfcRightCircular Cylinder	right_circular_ cylinder	TopoDS_Solid (BRepPrimAPI_MakeCylinder)	
			ellipsoid		
				TopoDS_Solid (BRepPrimAPI_MakeHalfSpace)	
				TopoDS_Solid (BRepPrimAPI_MakePrism)	
				TopoDS_Solid (BRepPrimAPI_MakeRevol)	
				TopoDS_Solid (BRepPrimAPI_Make Revolution)	
GM_Sphere	gml:Sphere	IfcSphere	sphere	TopoDS_Solid (BRepPrimAPI_MakeSphere)	
				TopoDS_Solid (BRepPrimAPI_MakePipe)	
				TopoDS_Solid (BRepPrimAPI_MakePipeShell)	
			torus	TopoDS_Solid (BRepPrimAPI_MakeTorus)	
			right_angular_wedge	TopoDS_Solid (BRepPrimAPI_MakeWedge)	
			tetrahedron		
			convex_hexahedron		
			faceted_primitive		

Boolean Geometry: Defined constructively, not by its boundary. Boolean geometry is the representation of 3D shapes by combining or subtracting simpler solids using Boolean operations like union, intersection, and difference.

Table 20

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
		IfcCsgSolid	Csg_solid		
		IfcBooleanResult	boolean_result	BRepAlgoAPI_BooleanOperation	

Solid/Shell: A solid is a topological object that encloses a region of space and a shell is a set of connected surfaces that form a boundary. A Solid is a closed shell.

Table 21

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
	gml:Abstract Solid				
GM_Solid	gml:Solid	IfcFacetedBrep	manifold_solid_brep	TopoDS_Solid (TopoDS_Face)	
		IfcFacetedBrepWithVoids			
		IfcAdvancedBrep	advanced_brep_shape_representation		
		IfcAdvancedBrepWithVoids			
		IfcSectionedSolidHorizontal			
		IfcFixedReferenceSweptAreaSolid			
		IfcDirectrixDerivedReferenceSweptAreaSolid			
		IfcExtrudedAreaSolid	extruded_area_solid		
		IfcExtrudedAreaSolidTapered			
		IfcRevolvedAreaSolid			
		IfcRevolvedAreaSolidTapered			

GM_Shell	gml:Shell				
				solid_replica	
				brep_2d	
		IfcSectionedSpine		sectioned_spine	
		IfcBoxedHalfSpace		boxed_half_space	
		IfcPolygonalBoundedHalfSpace			
				tesselated_solid	

Swept Surface:

A swept surface is a type of surface in geometric modeling created by moving a profile (curve) along a trajectory (path).

Table 22

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
		IfcSurfaceCurveSweptAreaSolid			
		IfcSweptDiskSolid	swept_disk_solid		
		IfcSweptDiskSolidPolygonal			
			swept_face_solid		
			swept_area_solid		
		IfcSurfaceOfLinearExtrusion	surface_of_linear_extrusion	Geom_SurfaceOfLinearExtrusion	
		IfcSurfaceOfRevolution	surface_of_revolution	Geom_SurfaceOfRevolution	
		IfcSweptSurface	surface_curve_swept_surface		
			fixed_reference_swept_surface		
			revolved_face_solid		

			extruded_face_solid		
			surface_curve_swept_area_solid		
			swept_disk_solid		

Volume:

Volume geometry is a way of representing a 3D object by directly defining the entire region of space it occupies, rather than just its boundary.

Table 23

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
			block_volume		
			pyramid_volume		
			eccentric_conical_volume		
			spherical_volume		
			ellipsoid_volume		
			cylindrical_volume		
			wedge_volume		
			tetrahedron_volume		
			hexadron_volume		
			toroidal_volume		
			bezier_volume		
			uniform_volume		
			quasi_uniform_volume		
			rational_b_spline_volume		
			trimmed_volume		

			locally_refined_spline_volume			
			rational_locally_refined_spline_volume			
			b_spline_volume			
			b_spline_volume_with_knots			

Axis/Placement:

Used to define orientation or alignment of objects, where to be placed and in what direction to extrude.

Table 24

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
		IfcAxis1Placement	axis1_placement	Geom_Axis1Placement	
		IfcAxis2Placement2D IfcAxis2Placement3D	axis2_placement_2d axis2_placement_3d	Geom_Axis2Placement	

Grids:

Table 25

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
	gml:RectifiedGrid				
	gml:RectifiedGridCoverage				
	gml:GridCoverage				

Operators:

Table 26

ISO 19107	GML	IFC	STEP	Open Cascade	WKT
			cartesian_ transformation_operator		

	IfcCartesianTransformationOperator3D	cartesian_ transformation_ operator_3d		
	IfcCartesianTransformationOperator2D IfcCartesianTransformationOperator2DnonUniform	cartesian_ transformation_ operator_2d	gp_Trsf2d	

Geometric complex/ composites:

Table 27

ISO 19107	GML	IFC	STEP	Open Cascade	WKT
GM_Complex	gml:Geometric Complex				
GM_Composite Point					
GM_Composite Curve	gml:Composite Curve	IfcComposite Curve	composite_ curv	composite_ curve	COMPOUNDCURVE (x y (z)(m))
GM_Composite Surface	gml:Composite Surface				
GM_Composite Solid	gml:Composite Solid				
			composite_ curve_segment		
			reparametrised_ composite_ curve_segment		
			rectangular_ composite_ surface		

Geometric Agggregates:

Table 28

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
GM_MultiPoint	gml:MultiPoint				MULTIPOINT
GM_MultiCurve	gml:MultiCurve				MULTILINESTRING
GM_MultiSurface	gml:MultiSurface				MULTIPOLYGON

GM_MultiSolid	gml:MultiSolid					
				IfcCartesianPointList2D		
				IfcCartesianPointList3D		

Interface Query:

Table 29

ISO 19107	GML	IFC	STEP	OpenCascade	WKT
boundary:Geometry					
convexHull:Geometry					
distance				gp_Circ::Distance() gp_Circ2d::Distance() gp_Lin::Distance(const gp_Pnt &theP) const gp_Lin::Distance(const gp_Lin &theOther) const gp_Lin2d::Distance(const gp_Pnt2d &theP) const gp_Lin2d::Distance(const gp_Lin2d &theOther) gp_Pln::Distance gp_Circ::Distance() etc	
buffer					
intersection				BRepClass_Intersector	
difference					
symDifference					
union					
contains				BRepClass3d_SolidClassifier	
crosses					
disjoint					
equals					
intersects					

overlaps	
touches	
within	
withinDistance	
relate	
3Dboundary:Geometry	
3DconvexHull:Geometry	
3Ddistance	
3Dbuffer	
3Dintersection	BRepClass3d_Intersector3d
3Ddifference	
3DsymDifference	
3Dunion	
3Dcontains	
3Dcrosses	
3Ddisjoint	
3Dequals	
3Dintersects	
3Doverlaps	
3Dtouches	
3Dwithin	
3DwithinDistance	

3Drelate

https://dev.opencascade.org/doc/refman/html/ https://old.opencascade.com/doc/occt-7.5.0/overview/html/index.html https://ifc43-docs.standards.buildingsmart.org/IFC/RELEASE/IFC4x3/HTML/lexical/IfcGeometricRepresentationItem.htm

https://www.web3d.org/documents/specifications/19775-1/V4.0/index.html

15

ANNEX C: OVERVIEW OF IFC GEOMETRY

15 ANNEX C: OVERVIEW OF IFC GEOMETRY

Table 30

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
Relational	Space boundaries (connection geometry)	12.6% (1676)	`IfcRelSpaceBoundary`: Surfaces representing adjacencies between spaces and building elements primarily intended for thermal simulation.	`geof:sfTouches`
Relational	Interferences	1.4% (3)	`IfcRelInterferesElements`: Primarily intended for modelling clash detection outcomes and over- and underpasses in infrastructure models. Likely has seen little use in practice.	`ST_Intersects`
Relational	Metric constraints	0%	`IfcMetric`: distances, parametric expressions between properties and geometry; supported in the schema (has limitations on indexing into unordered aggregates) and rarely used in practice.	`ST_Distance` `owl:sameAs` `rdfs:seeAlso`
Relational	Clearance geometry	0%	`IfcShapeRepresentation[Repr 'Clearance']`: Dedicated shapes encoding minimum clearances (e.g., around equipment, maintenance envelopes, door swings); a separate representation with its own context	
Product	Relative positioning	97.6% (1298)	`IfcLocalPlacement` provides hierarchical local coordinate systems for products. Informal proposition are in place to define a placement hierarchy that follows the project decomposition structure	gml:AffineCS
Product	Linear positioning	4.8% (527)	`IfcLinearPlacement` provides linear positioning that derives an orientation and positioning frame	ISO 19148:2021

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
			from a length measure along an (IfcAlignment) curve	
Product	Grid placement	0%	`IfcGridPlacement` provides placement by means of the computed intersection between two grid axes with underlying curve geometry	
Product	Openings	49.1% (168)	`IfcRelVoidsElements` + `IfcOpeningElement`/ `IfcVoidingFeature` models a subtractive boolean operation applied to host elements (walls, slabs, etc.); in principle extensible to other feature kinds such as `IfcProjectionElement`. In IFC2X3 CoordinationView these were to be processed by the importer. In IFC4 Reference View, in principle, these are precomputed at export time.	`ST_Difference`
Product	Material association	71% (324)	`IfcRelAssociatesMaterial`, layered / list / constituent / profile-based Links product geometry to material models: layered elements, material lists, constituents, profile-based materials. Especially layered materials (IfcMaterialLayer Set[Usage]) have geometric implications because it semantically slices the geometry by means of planes defined from offsets along a reference curve (for vertical elements) or along the extrusion direction.	
Product	Parametric geometry as geometry-related semantics via property sets (e.g. `Pset_ MechanicalFastener` `IfcWindowLiningPrc` `Pset_ ProvisionForVoid`,	29% (73)	Semantic parameters that can be used to drive or reconstruct geometry when explicit representations are absent.	

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
	most quantities `Qto_*`)			
Product	Product-level composite curves defining the 'business-logic' of parametric alignment segment nesting within a curve	4.2% (504)	Decomposition of IfcAlignment into IfcAlignmentSegment with parametric geometry (typically redundant with item-level curves).	OGC LandInfra/InfraGML alignment model
Product	Aggregation of product into assemblies: composition (`IfcElementAssembl` nesting (ports, alignment segments)	96.9% (29)	Product-level aggregation (e.g., ports, sub-elements, segments)	
Item	Basic 0D/1D/2D items: points, curves, faces (IfcCartesian Point, IfcPolyline, Ifc CompositeCurve, Ifc Face)	99.2% (34194)	Simple representations or underlying points, polylines, composite curves (including circular/arc segments), and faces used in higher-level geometry.	SFA: Point, LineString, MultiLine String, Polygon; GML: gml:Point, gml:Curve, gml:LineString
Item	Profiles: parametric (e.g I shape) / arbitrary / derived / composite (IfcProfile Def, IfcArbitrary ProfileDef, Ifc DerivedProfileDef, Ifc CompositeProfileDef) [Note: not strictly an independent Ifc RepresentationItem]	76.9% (763)	Cross-section definitions used by sweeps and structural members; parametric catalog profiles and arbitrary section shapes.	
Item	Annotation geometry: text literals, hatches and fills	11.6% (101)	`IfcTextLiteral`, `IfcFillAreaStyle` etc. 2D annotation geometry, symbolization and fills; mostly for documentation, not primary 3D model.	
Item	Explicit (BRep) representations of 3D **faceted** solids and surfaces	43.9% (876)	`IfcFacetedBrep`	SFA: PolyhedralSurface

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
Item	Explicit (BRep) representations of 3D **curved** solids and surfaces	6.5% (112)	Advanced BRep: with (e.g) NURBS faces (IfcAdvancedFace)	
Item	Explicit mesh/ polyhedron representations (called tessellation)	12.6% (163)	`IfcTessellatedFaceSet` `IfcTriangulatedFaceSet` `IfcPolygonalFaceSet` Indexed face sets representing triangulated or polygonal surfaces/solids; compact storage via point lists and indices.	SFA: PolyhedralSurface gml: TriangulatedSurface
Item	TIN / surface models	2% (1)	`IfcTriangulatedIrregularNet` Specialized tessellated representation for terrain and horizontal surfaces (single f(X, Y)→Z for any X,Y); typical for infrastructure and terrain models.	gml:Tin
Item	Swept / procedural solids: extrusion, revolution, sweeps along curve (disk, arbitrary, fixed reference), sectioned solids and surfaces	75.5% (703)	3D solids defined by sweeping profiles along vectors or curves: straight/linear extrusions, revolutions, sweeps along curves (swept disk, arbitrary orientation, fixed reference), tapered forms, sectioned solids/surfaces. Note that this is a fairly large category with many items and combinatorial possibilities lumped into one.	
Item	Axis geometry	47.6% (321)	Bounded curve geometries as (often) a secondary representation describing the longitudinal direction of a solid or element. Material layer sets are defined relative to this axis.	
Item	Boolean operations (CSG trees)	30.6% (122)	Solids defined as binary trees combining operands via union, intersection, difference; often nested and require recursive traversal.	SFA: ST_Union, ST_Intersection, ST_Difference
Item	CSG primitives	1.7% (8)	`IfcCsgPrimitive3D` Are only part of the CSG expressions, but include things like cylinder and sphere which are 'harder'	

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
			to produce otherwise (as revolutions)	
Item	Surface-only geometry for connection / space boundaries and advanced face surfaces (extrusion, revolution, bounded, NURBS)	31.8% (1521)	Surfaces used without explicit solid closure, e.g. for space boundaries or analytical/ architectural surfaces.	gml:Surface
Item	Styling by item as a whole (IfcStyledItem, IfcSurfaceStyle, Ifc CurveStyle)	68.3% (1458)	Assignment of colors, line styles to geometric items or tessellated faces as a whole	
Item	Granular styling by sub-item (IfcSurface Style, IfcIndexed ColourMap)	-	E.g the application of a different color to a face within a shell, or per-face colors in tesselations using IfcIndexedColourMap, or retaining distinct surface styles after applying boolean operations	
Item	Styling by item: Textures	0.8% (2)	`IfcSurfaceStyleWithTextures` and three subtypes of `IfcSurfaceTexture` for a specialization of the above with texture as a binary blob, a compressed inline literal or an external URI reference. The complexity of supporting textures is mostly related to handling and defining texture coordinates (UV) on the procedural geometries (such as sweeps and booleans). Most likely `IfcReparametrisedCompositeC` is also intended for supporting textures so that the U space of the profile prior to extrusion can be controlled.	
Item	Mapped items / reusable definitions	46.3 1247	`IfcMappedItem` and type object occurrences: Instancing and transformations: one definition, multiple placed occurrences with transforms. In IFC2X3 it was not allowed to re-use the full	

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
			representation as a whole, later in IFC4 that is possible.	
Item	(Non-)uniform scaling and mirroring	-	`IfcMappedItem` is also the only mechanism in IFC that supports scaling and mirroring.	
Item	Composite representations (Ifc GeometricSet, Ifc CompositeProfile Def, multiple shape representation items)	51.4% (231)	Sets of curves, surfaces, and points combined into a single representation; multi-rep shape representations for one object.	gml:MultiCurve, gml:Multi Surface, gml:CompositeCurve, gml:CompositeSurface
Item	Topological representations for analytical structures	0.3% (760)	`IfcEdge` `IfcVertex` `IfcFace` `IfcTopologyRepresentation` Explicit topology (vertices, edges, faces) as top-level items; often used in structural/analytical models.	
Item	Point clouds and coverages	0.2% (1)	`IfcShapeRepresentation` + `IfcCartesianPointList3D`	
Context	Representation context (IfcGeometric Representation Context): identifier, target scale, clearance geometries, subcontexts	-	Defines modeling context: dimensionality, precision, world coordinate system, true north, representation identifiers; may host clearance contexts.	
Context	Tolerance and precision settings	-	`IfcGeometricRepresentationC Precision` Model-level precision and tolerances that influence how geometry is interpreted and when elements are considered coincident. Only linear tolerance, no angular setting and no local (e. g per-vertex) possibilities.	
Georeferen	Local engineering CRS (IfcGeometric Representation Context.World CoordinateSystem, true north)	-	Defines the local engineering coordinate system	

LEVEL	CONCEPT	PREVALEN	NOTES	RELATED TERMS FROM ISO / OGC (SFA, GEOSPARQL, GML, ...)
Georeferen	Map/projection-based georeferencing incl. vertical reference	12.4% (2)	`IfcMapConversion`, `IfcProjectedCRS` Conversion from local model coordinates to a map/projected CRS; explicit link to EPSG or other CRS identifiers; since IFC4/IFC4x3.	ISO 19111 referencing by coordinates and transformations; SFA: SRID on geometries.
Georeferen	Postal address	38.6% (3)	`IfcPostalAddress`, `Pset_Address`	
Georeferen	IfcSite reference coordinates	-	`IfcSite` has DMS reference coords for latitude and longitude	

15.1. Methodology for testing prevalence.

Testing the prevalence of entities in existing sample models is often criticized because it is hard to find a body of representative IFC models (every tool and vendor cater to their own niche) and prevalence does not equate value. Lastly, per definition, an approach like this is backwards looking and not forward looking. At the same time, the investigation into 3D GeoSPARQL is explicitly use-case driven.

A random sample of 1_000 files (dataset 1) is taken from a collection of models for which it is allowed to publish in aggregate form. For novel concepts such as (IfcLinearPlacement, IfcRelInterferes, IfcAlignmentSegment, ...) a 500 file sample of IFC4.3 files is selected (dataset 2; because the amount of models for older schema versions is larger). It is possible that there are invalid or partial files in this sample, there is no verdict on quality, correctness or applicability of these files. Example files from the specification can have a distorting impact on the sample, also there is no filtering for (near-) duplicates. Some values such as IfcSite.RefLatitude -Longitude are well known to contain placeholder values on older exports such as based on the headquarters of the software vendor.

For performance reasons we don't use an IFC parser for testing the prevalence of entities. Therefore, in theory it is possible that we match fragments of comments or human readable text in the model, but this is given the patterns highly unlikely. We use ripgrep `rg` to query the model as text files on disk. Due to the text-based parsing we cannot fully asses the context in which entities are used — in particular entities like IfcRelAggregates are discussed here in an building-element context, but are also found in the spatial structure and therefore the prevalence is reported higher.

Table 31

QUERY STRING	DATASET INDEX	% OF MODELS	AVG NR INSTANCES WHEN USED
IFCRELAGGREGATES	1	96.9	29
IFCCARTESIANPOINT	1	99.2	34194
PROFILEDEF	1	76.9	763
IFCANNOTATIONFILLAREA| IFCANNOTATIONSURFACE| IFCDRAUGHTINGCALLOUT| IFCFILLAREASTYLEHATCHING|IFCTEXTLITERAL	1	11.6	101
IFCFACETEDBREP	1	43.9	876
IFCADVANCEDBREP	1	6.5	112
IFCTESSELLATEDFACESET| IFCTRIANGULATEDFACESET|IFCPOLYGONALFACESET	1	12.6	163
IFCTRIANGULATEDIRREGULARNETWORK	2	2	1
IFCEXTRUDEDAREASOLID|IFCREVOLVEDAREASOLID| IFCFIXEDREFERENCESWEPTAREASOLID| IFCSURFACECURVESWEPTAREASOLID| IFCDIRECTRIXDERIVEDREFERENCESWEPTAREASOLID	1	75.5	703
IFCSHAPEREPRESENTATION.*'Axis	1	47.6	321
IFCBOOLEANRESULT|IFCBOOLEANCLIPPINGRESULT	1	30.6	122
IFCBLOCK|IFCRECTANGULARPYRAMID| IFCRIGHTCIRCULARCONE| IFCRIGHTCIRCULARCYLINDER|IFCSPHERE	1	1.7	8
IFCCURVEBOUNDEDPLANE| IFCCURVEBOUNDEDSURFACE|IFCBSPLINESURFACE| IFCCYLINDRICALSURFACE|IFCPLANE\(| IFCSPHERICALSURFACE|IFCTOROIDALSURFACE| IFCSURFACEOFLINEAREXTRUSION| IFCSURFACEOFREVOLUTION	1	31.8	1521
IFCSTYLEDITEM|IFCSURFACESTYLE|IFCCURVESTYLE	1	68.3	1458
-			

QUERY STRING	DATASET INDEX	% OF MODELS	AVG NR INSTANCES WHEN USED
IFCBLOBTEXTURE|IFCIMAGETEXTURE| IFCPIXELTEXTURE	1	0.8	2
IFCMAPPEDITEM	1	46.3	1247
-			
IFCGEOMETRICSET|IFCCOMPOSITEPROFILEDEF| IFCGEOMETRICCURVESET| IFCSHAPEREPRESENTATION\(.\(.,	1	51.4	231
IFCTOPOLOGYREPRESENTATION	1	0.3	760
IFCSHAPEREPRESENTATION.*'Point	1	0.2	1
-			
-			
-			
IFCPROJECTEDCRS|IFCGEOGRAPHICCRS| IFCMAPCONVERSION|IFCRIGIDOPERATION	2	12.4	2
IFCPOSTALADDRESS|Pset_Address	1	38.6	3
-			

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ANNEX O: HISTORY

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BIBLIOGRAPHY

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